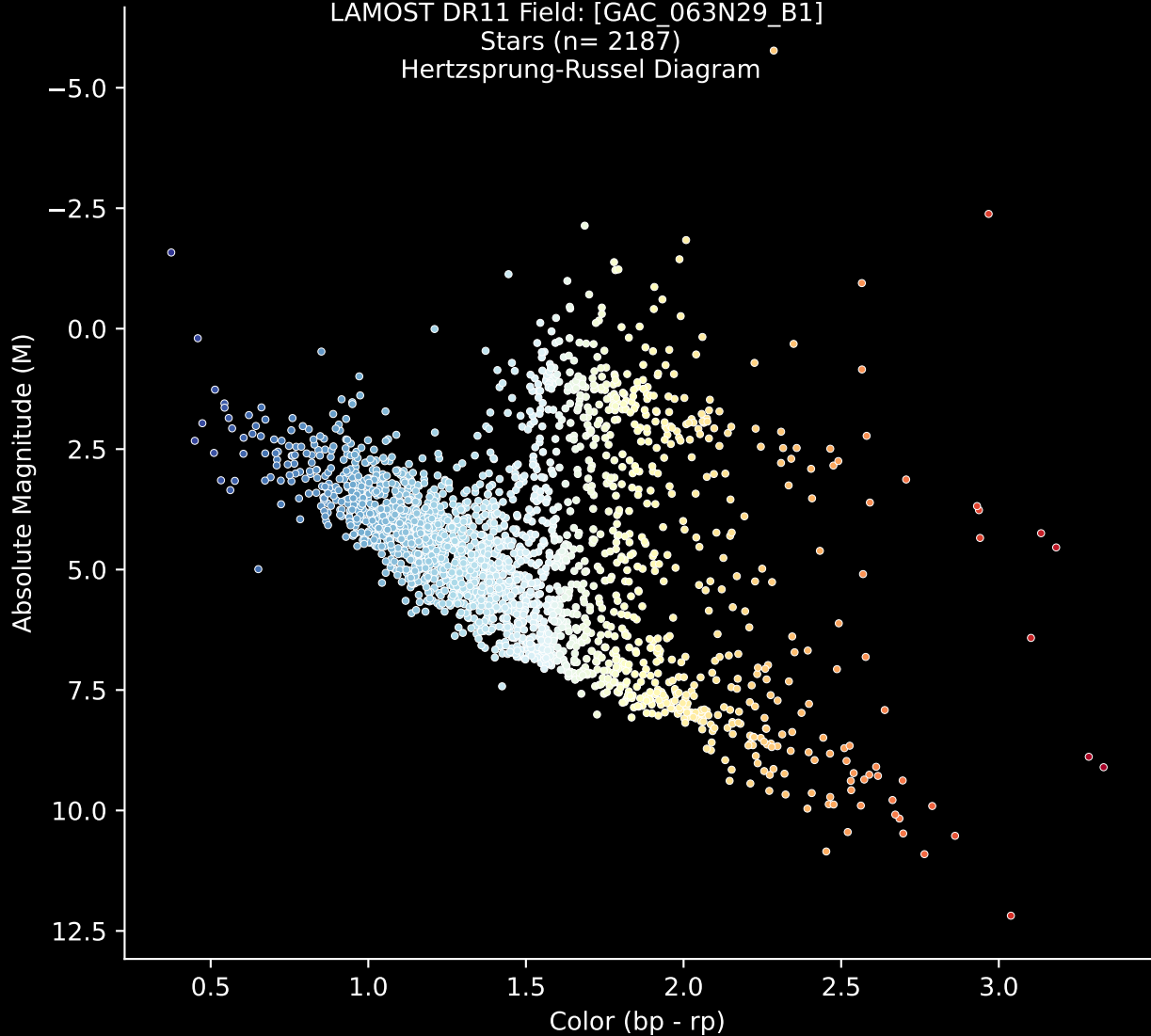


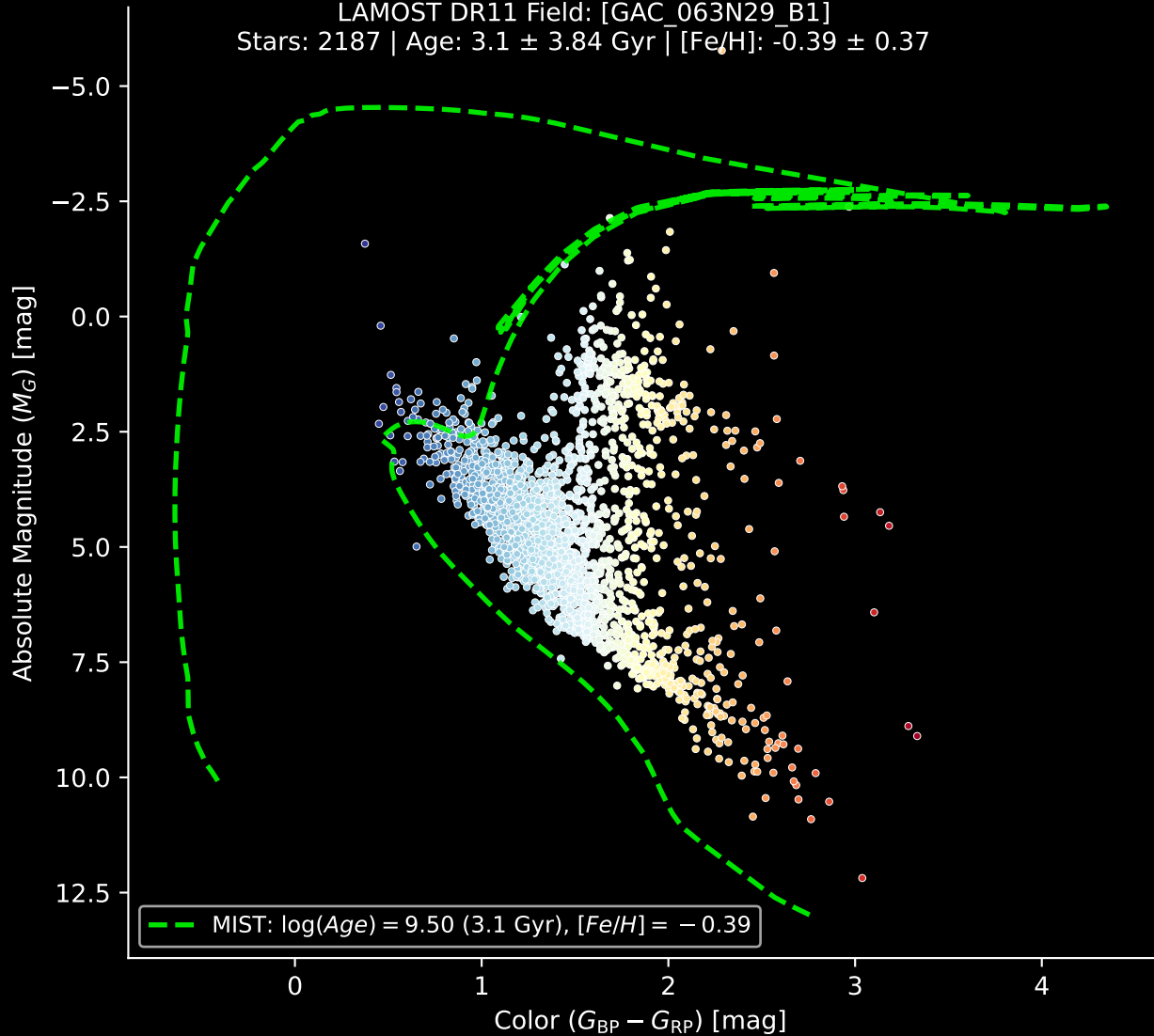
LAMOST DR11 Field: [GAC_063N29_B1]

Stars (n= 2187)

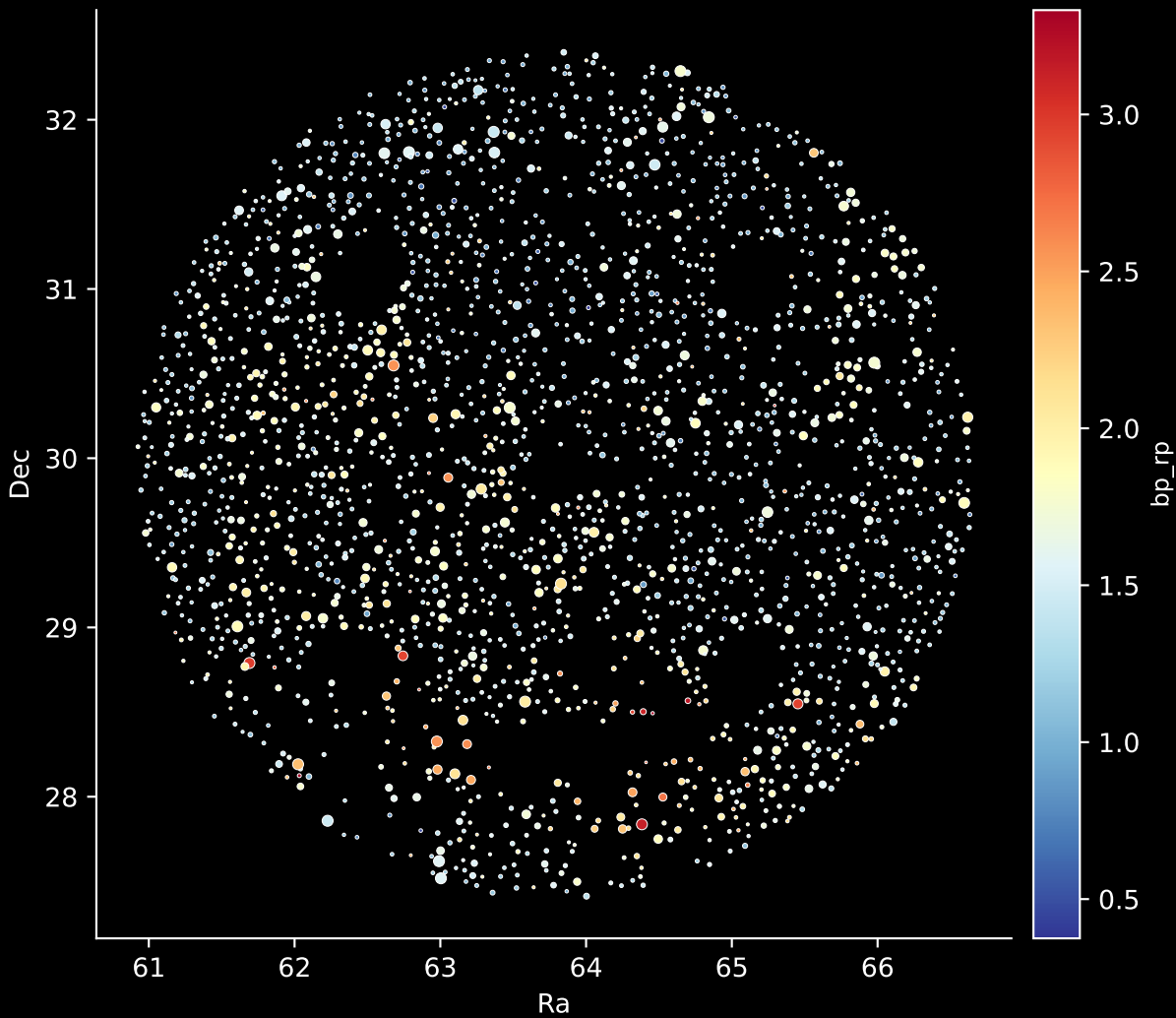
Hertzprung-Russel Diagram



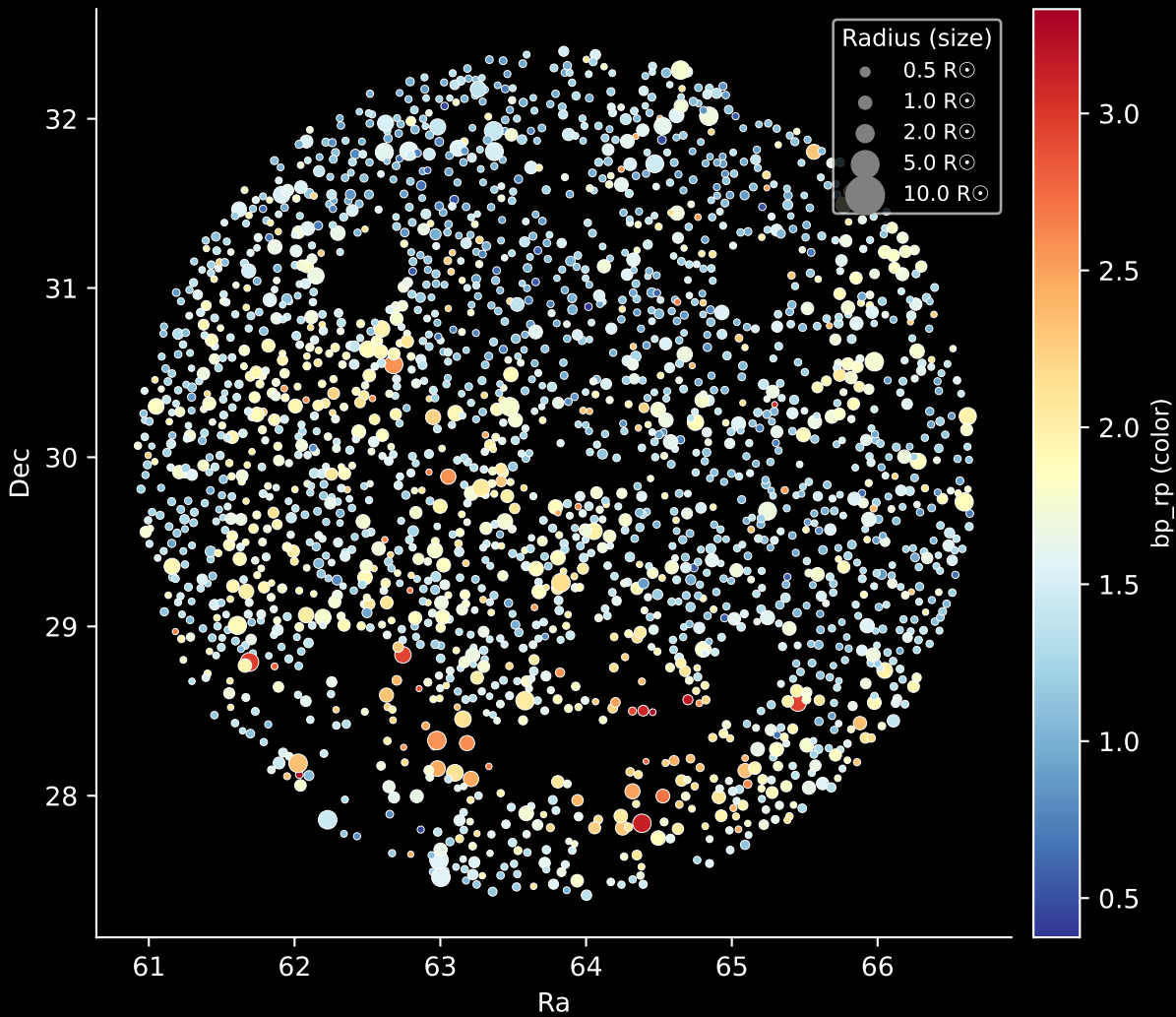
LAMOST DR11 Field: [GAC_063N29_B1]
Stars: 2187 | Age: 3.1 ± 3.84 Gyr | $[Fe/H]: -0.39 \pm 0.37$



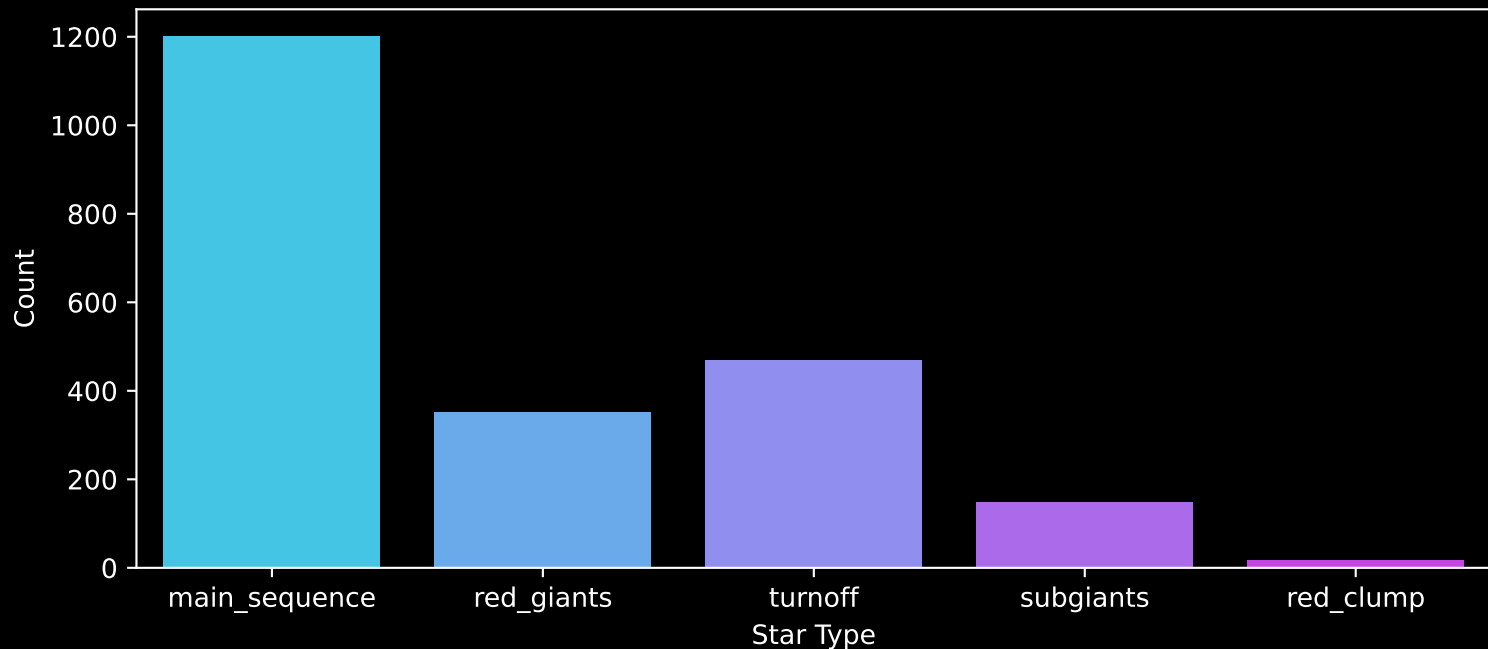
LAMOST DR11 Field: [GAC_063N29_B1]
Stars: 2187



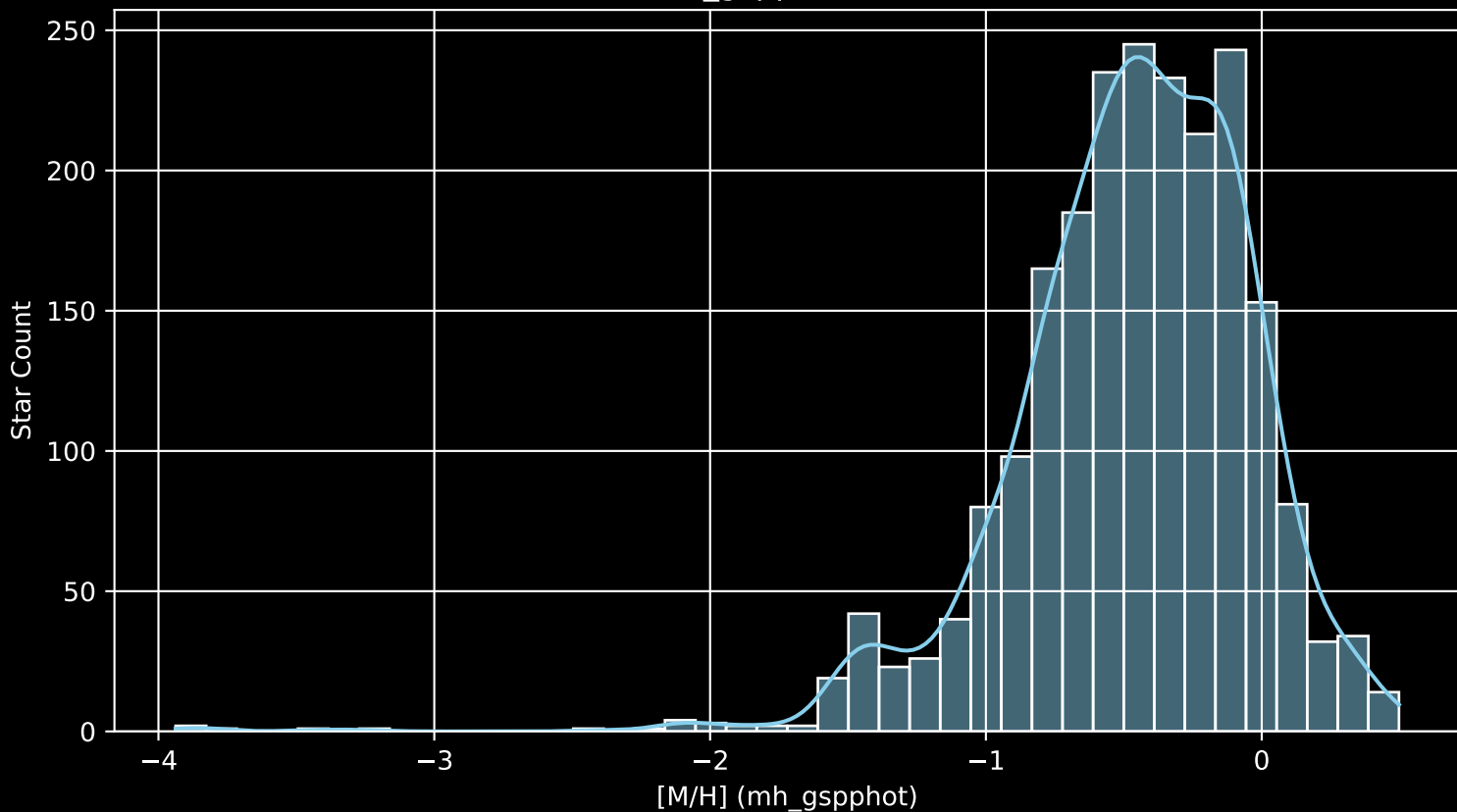
LAMOST DR11 Field: [GAC_063N29_B1]
Stars: 2187



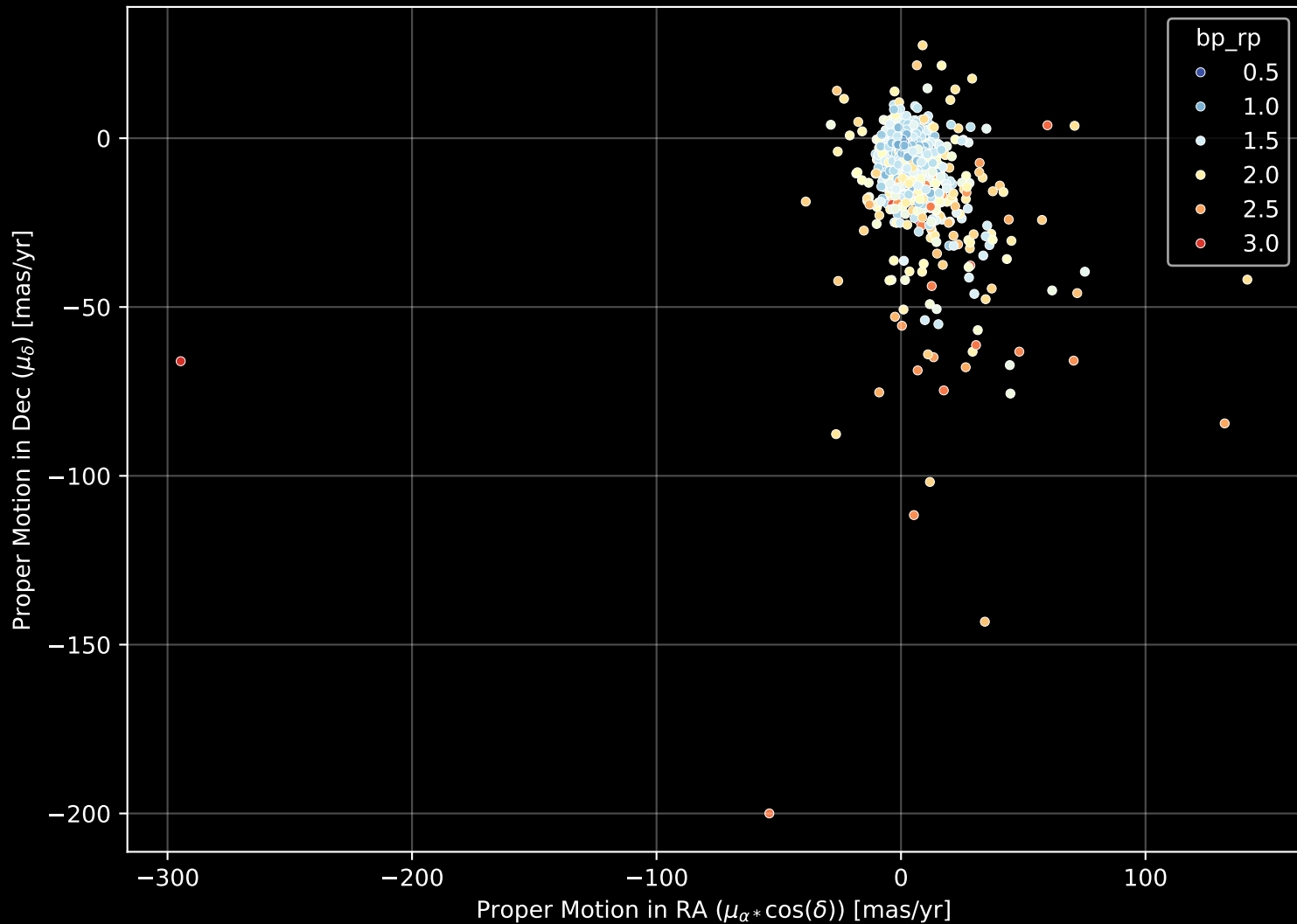
LAMOST DR11 Field: [GAC_063N29_B1]
Count plot of Star Type



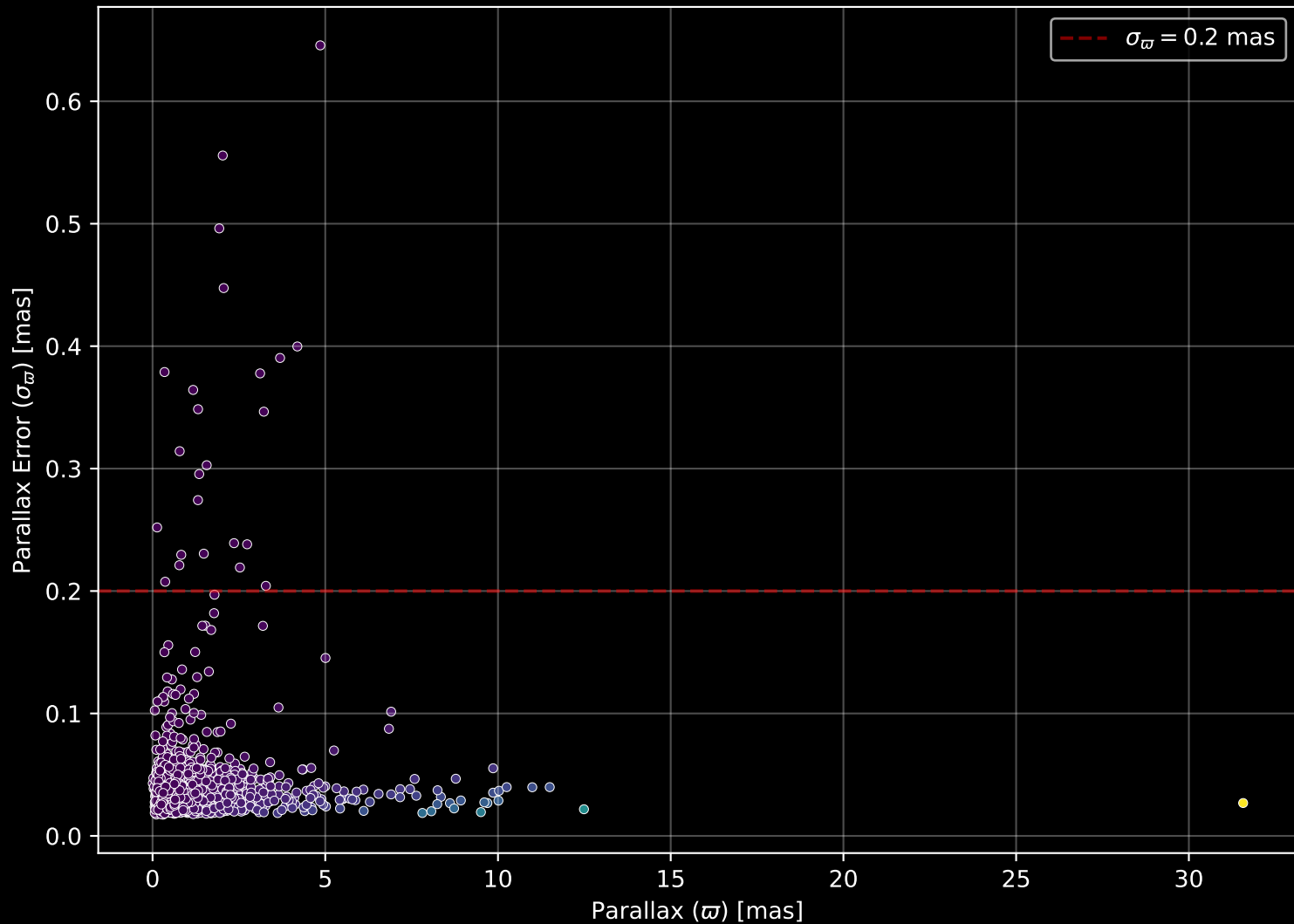
LAMOST DR11 Field: [GAC_063N29_B1
[M/H] (mh_gspphot) (n= 2181)



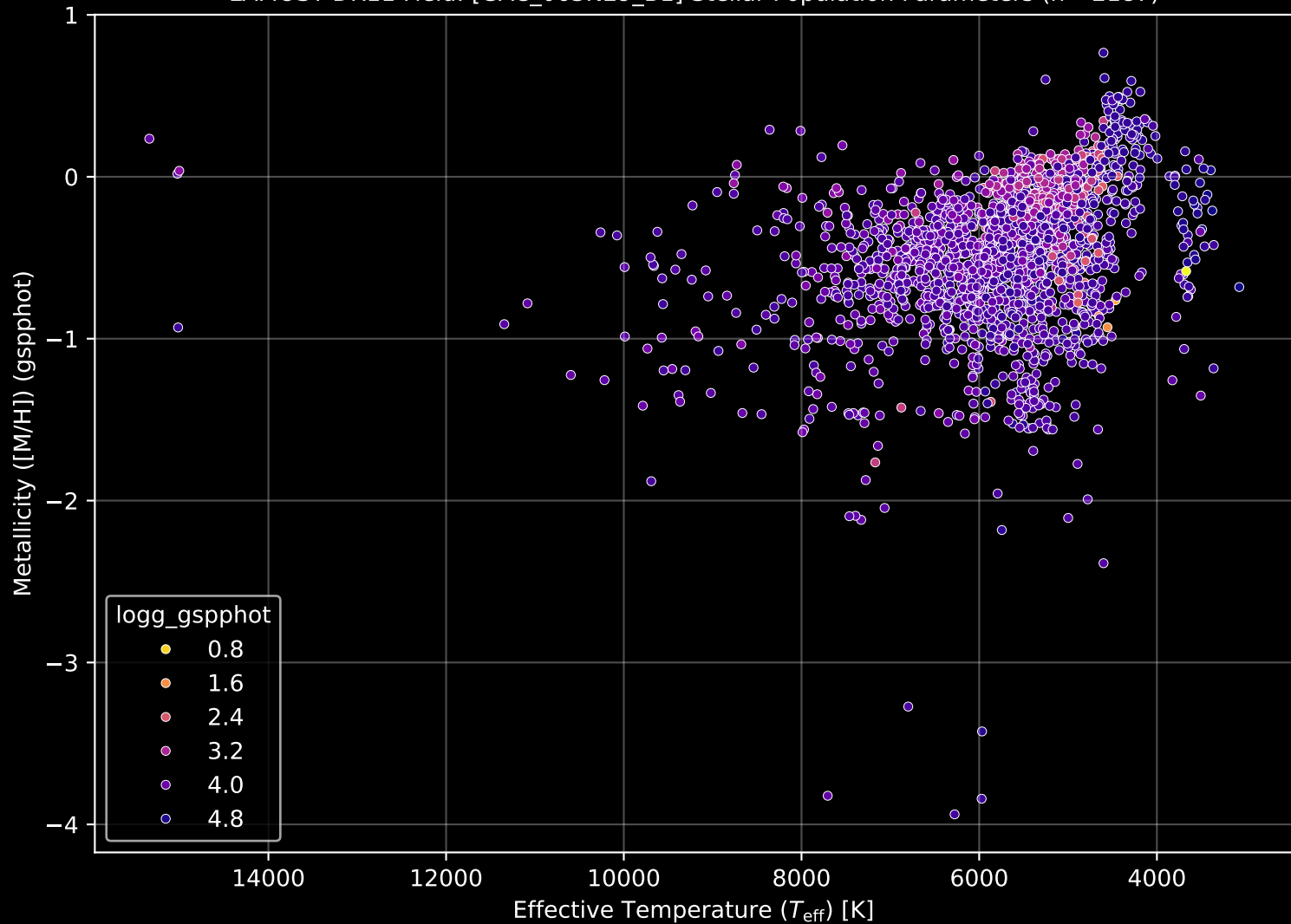
LAMOST DR11 Field: [GAC_063N29_B1] Proper Motion Diagram (n= 2187)



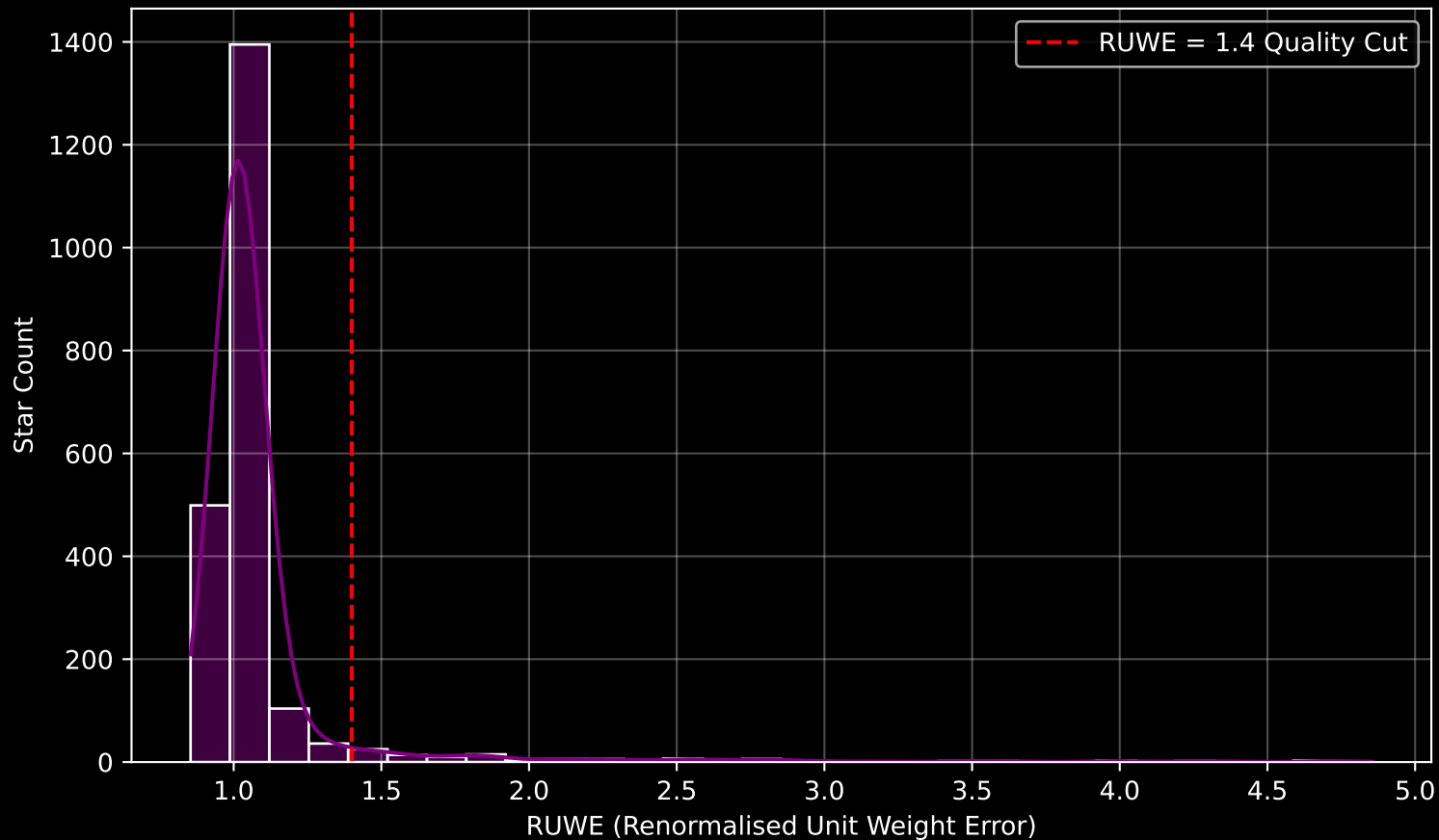
LAMOST DR11 Field: [GAC_063N29_B1] Parallax Quality (n= 2187)



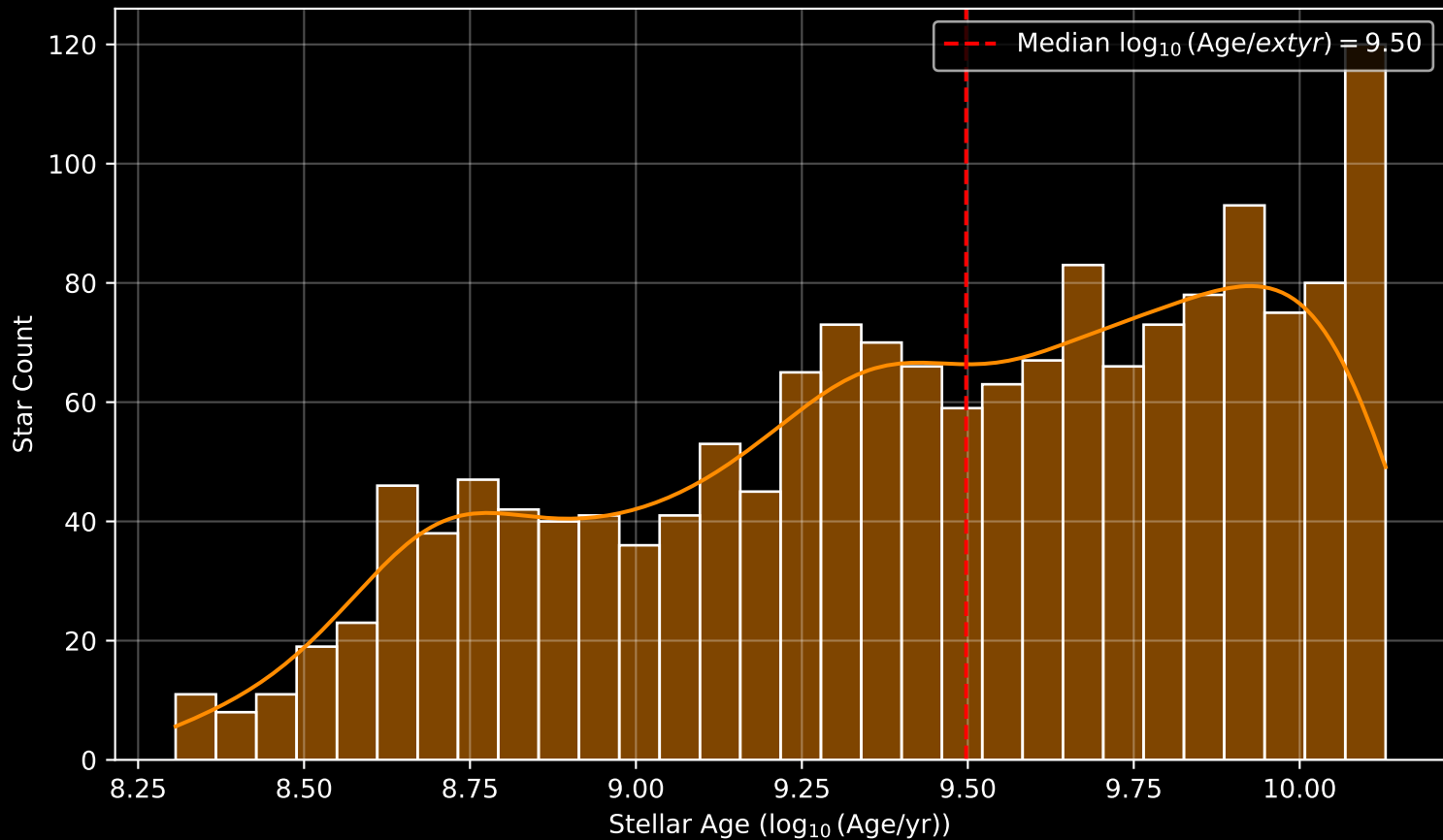
LAMOST DR11 Field: [GAC_063N29_B1] Stellar Population Parameters (n= 2187)



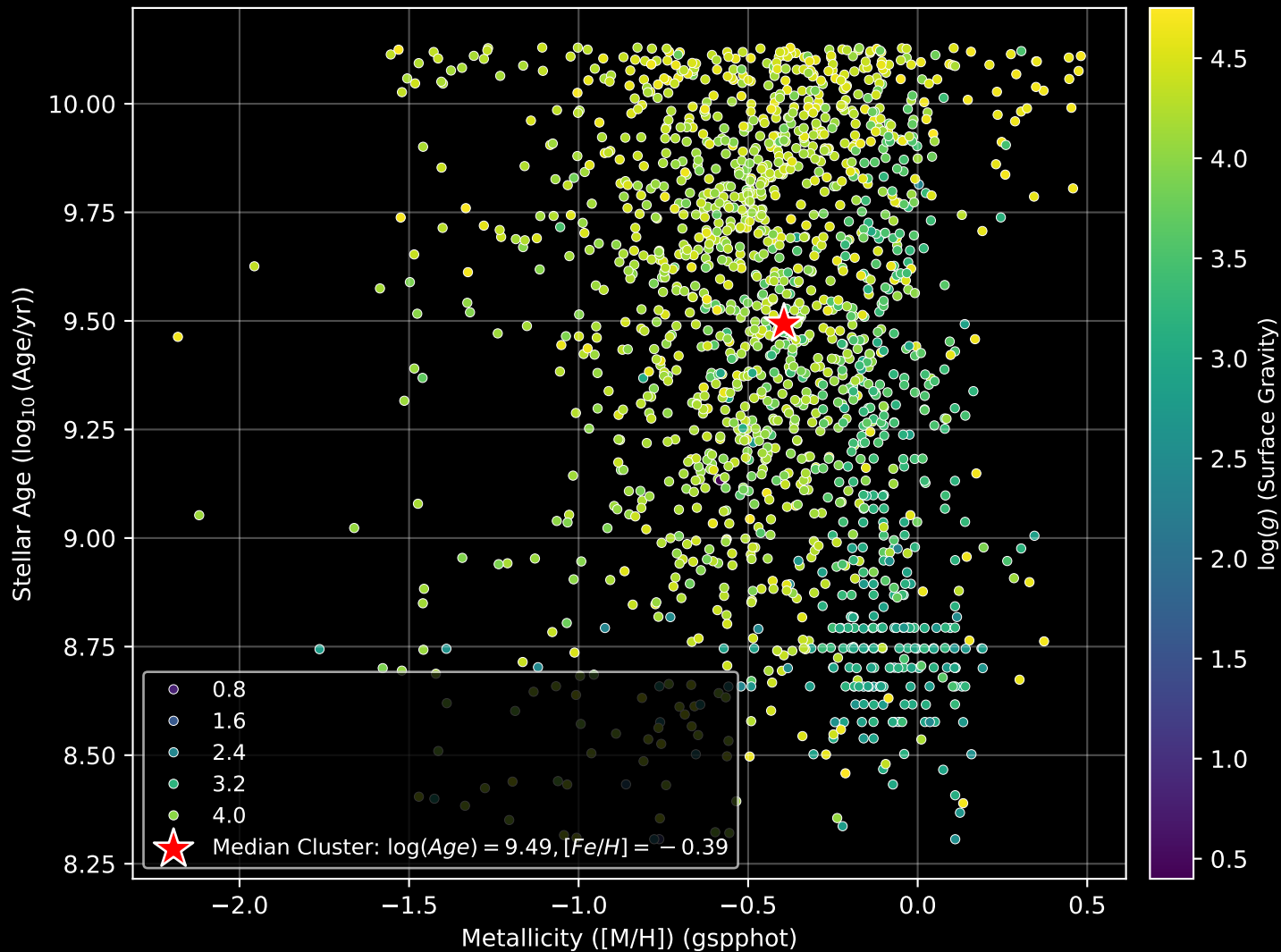
LAMOST DR11 Field [GAC_063N29_B1] RUWE Distribution (n= 2154)



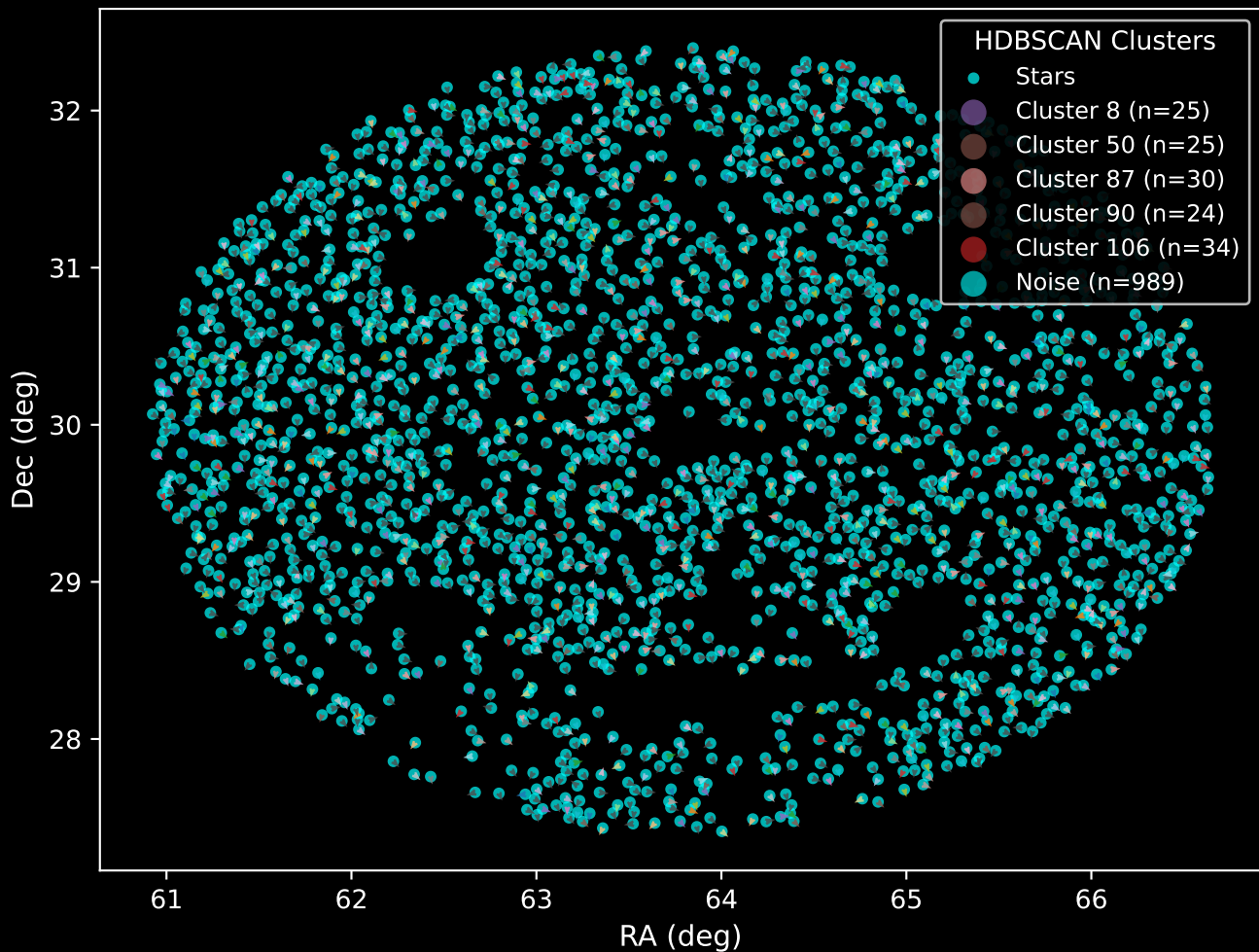
LAMOST DR11 Field: [GAC_063N29_B1] Age Distribution (n= 1632)



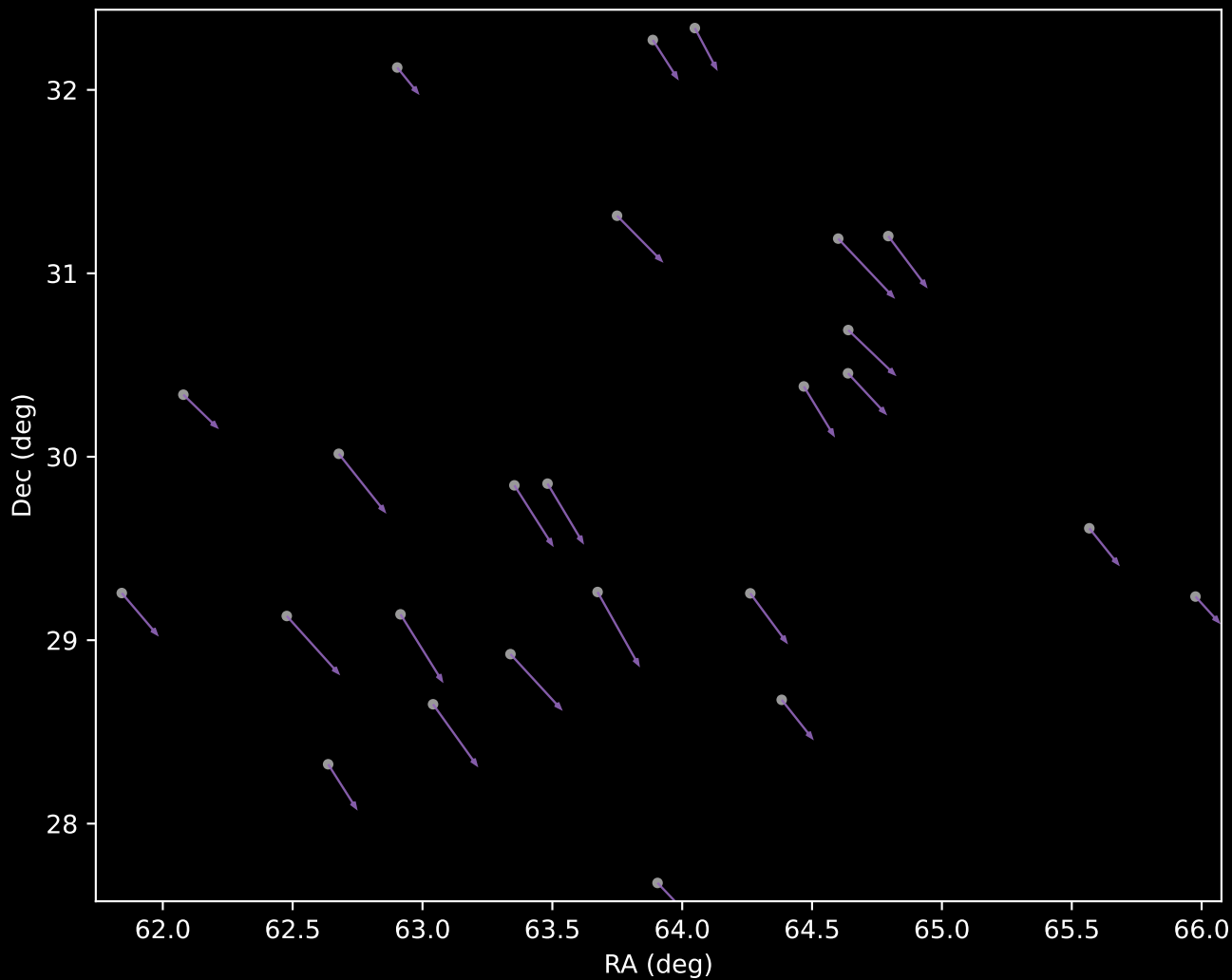
LAMOST DR11 Field: [GAC_063N29_B1] Age-Metallicity Relation (n= 1628)



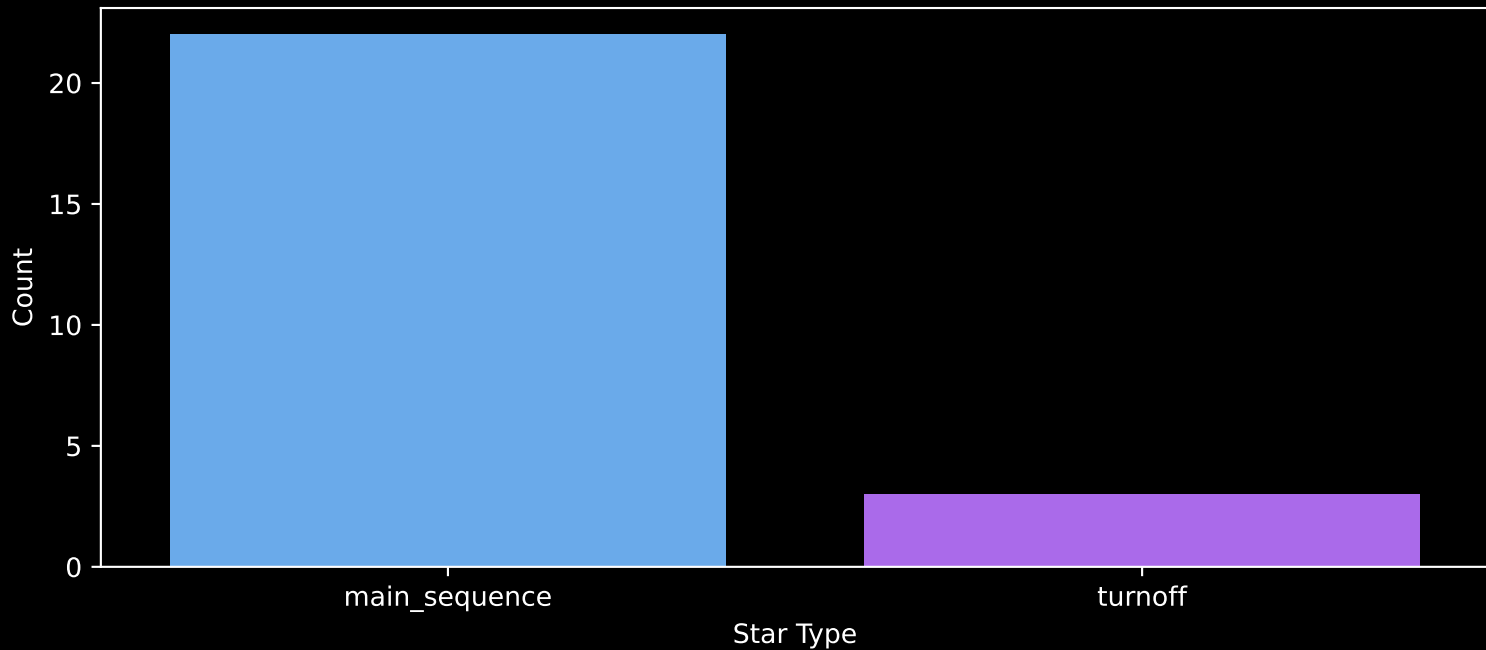
LAMOST DR11 Field [GAC_063N29_B1]
Proper Motion Vectors with HDBSCAN
(n=2187)



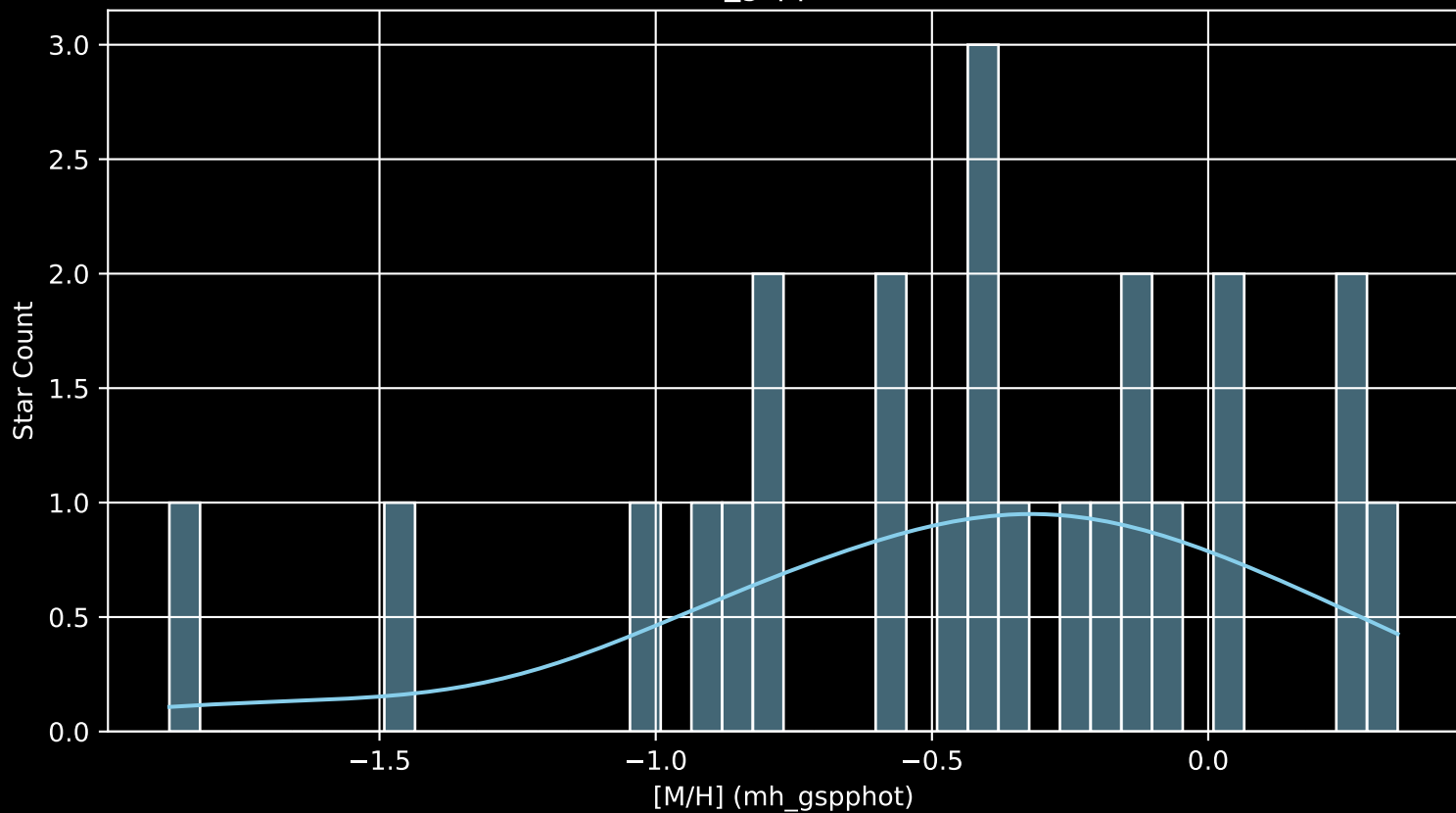
LAMOST DR11 Cluster 8 Proper Motion Vectors
(n=25)



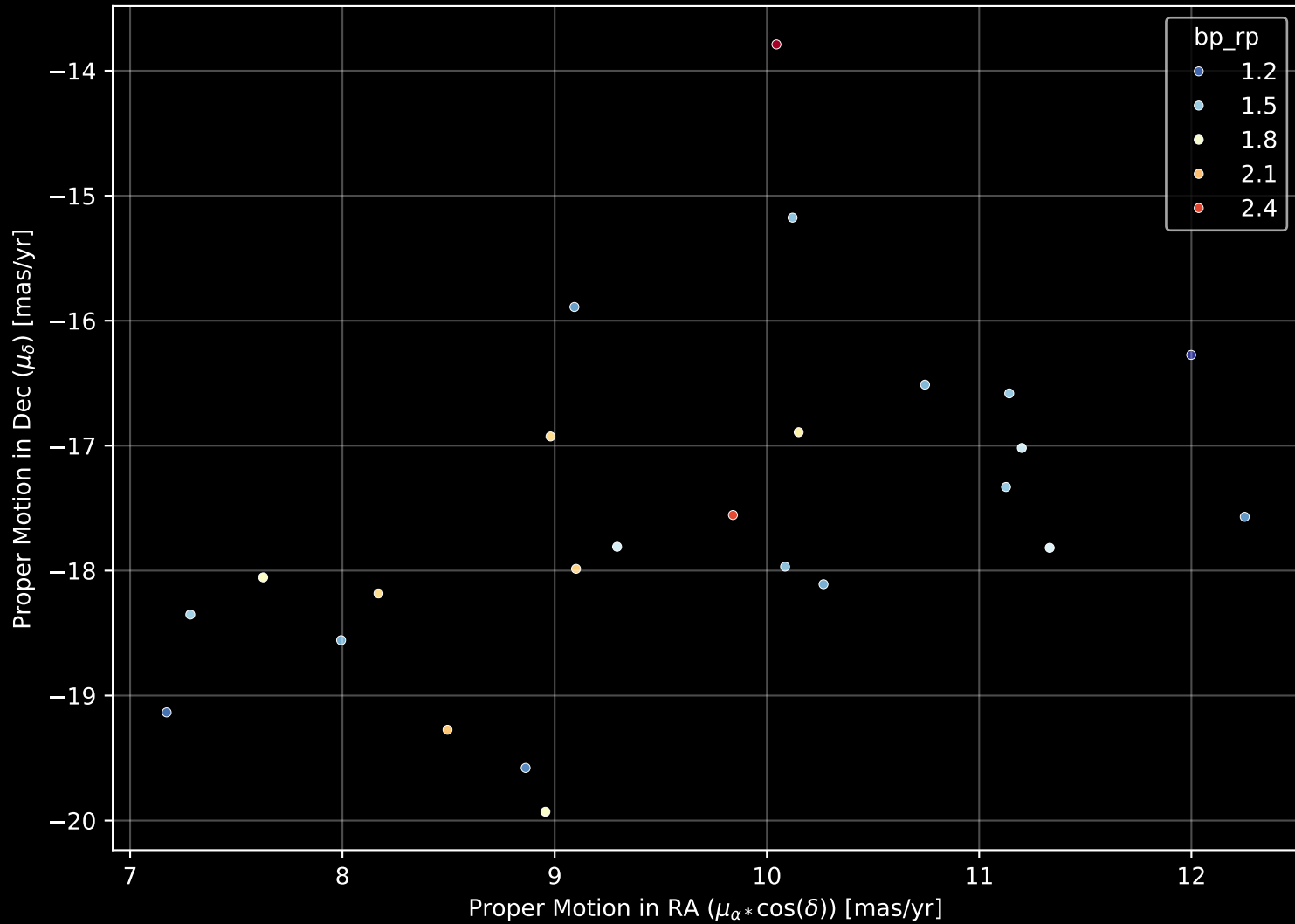
Cluster 8
Count plot of Star Type (n=25)



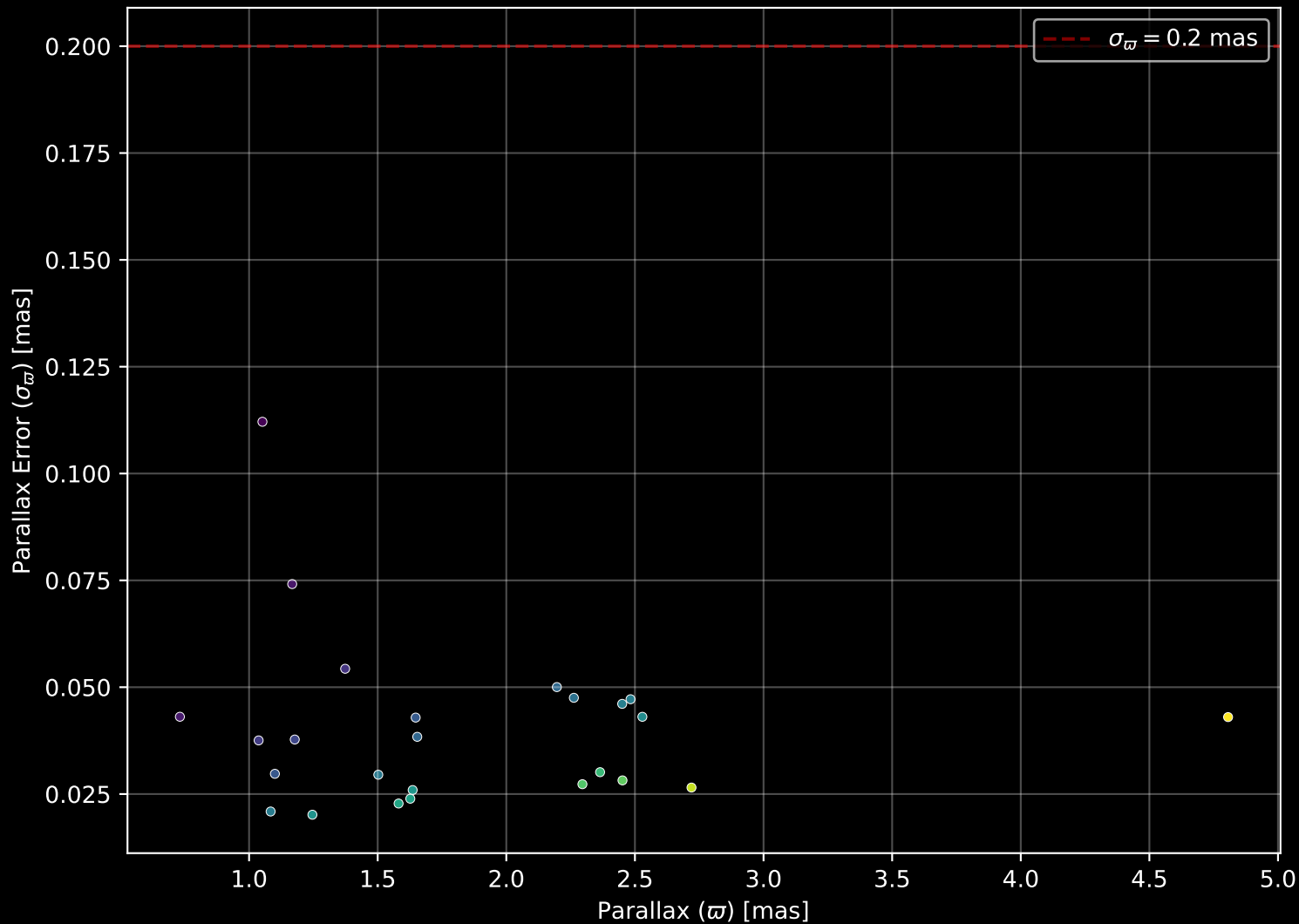
Cluster 8
[M/H] (mh_gspphot) (n=24)



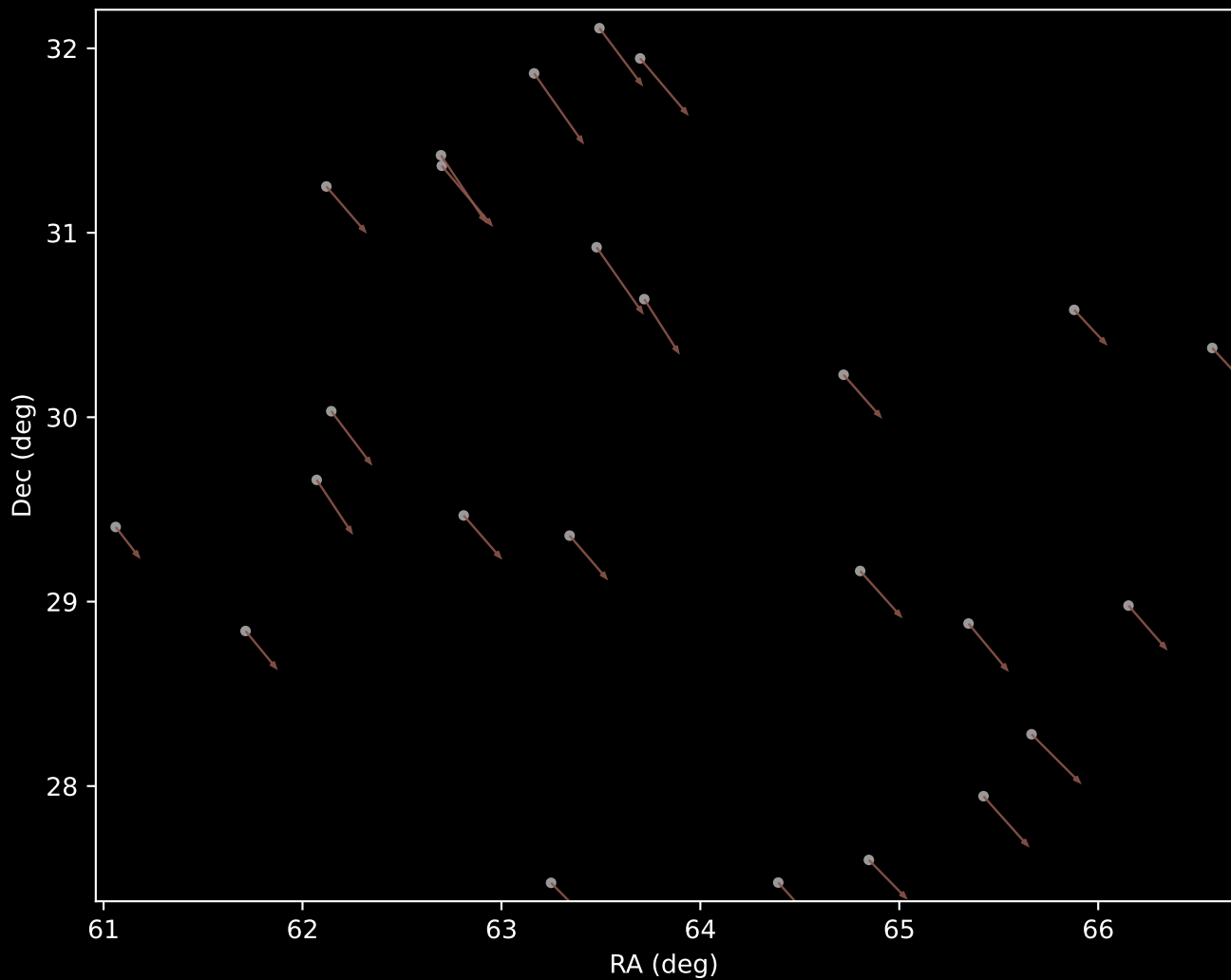
Cluster 8 Proper Motion Diagram (n=25)



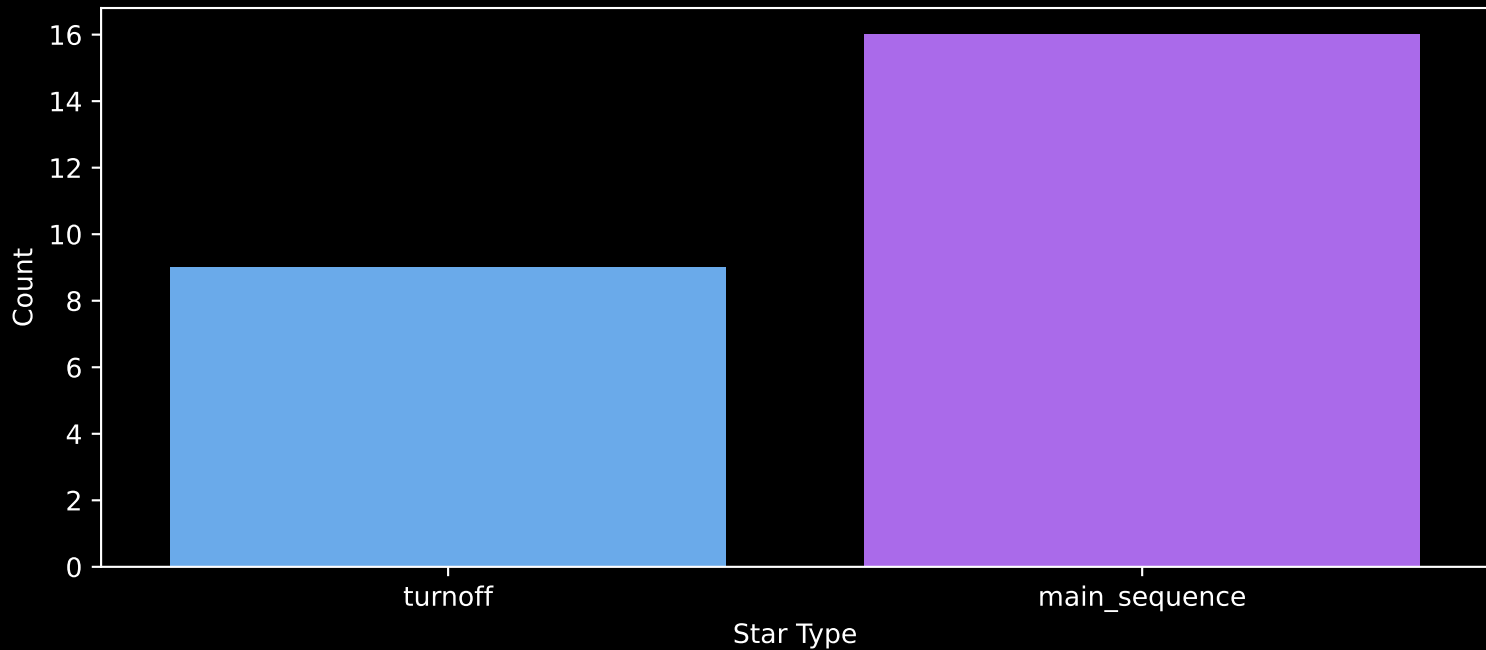
Cluster 8 Parallax Quality (n=25)



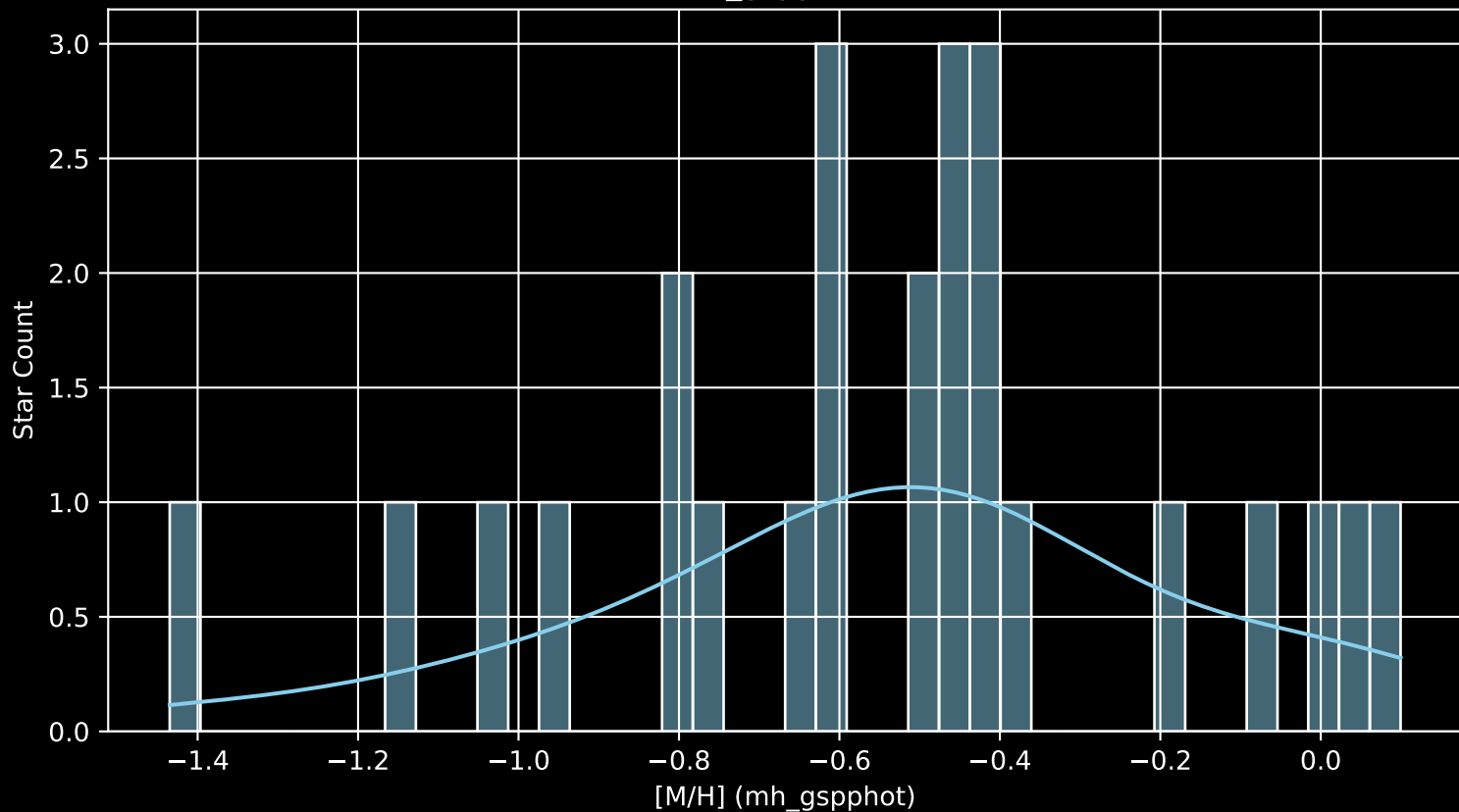
LAMOST DR11 Cluster 50 Proper Motion Vectors
(n=25)



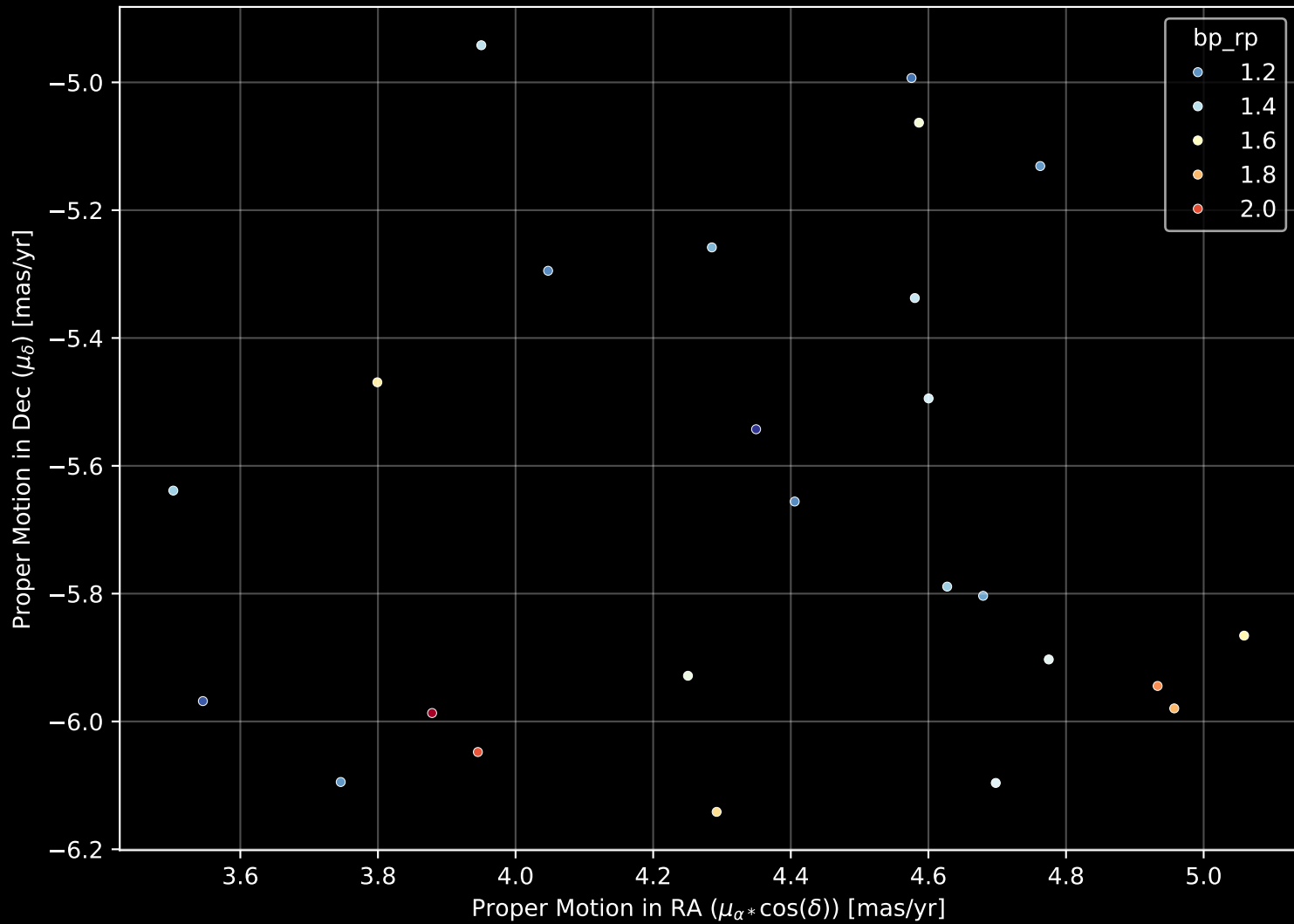
Cluster 50
Count plot of Star Type (n=25)



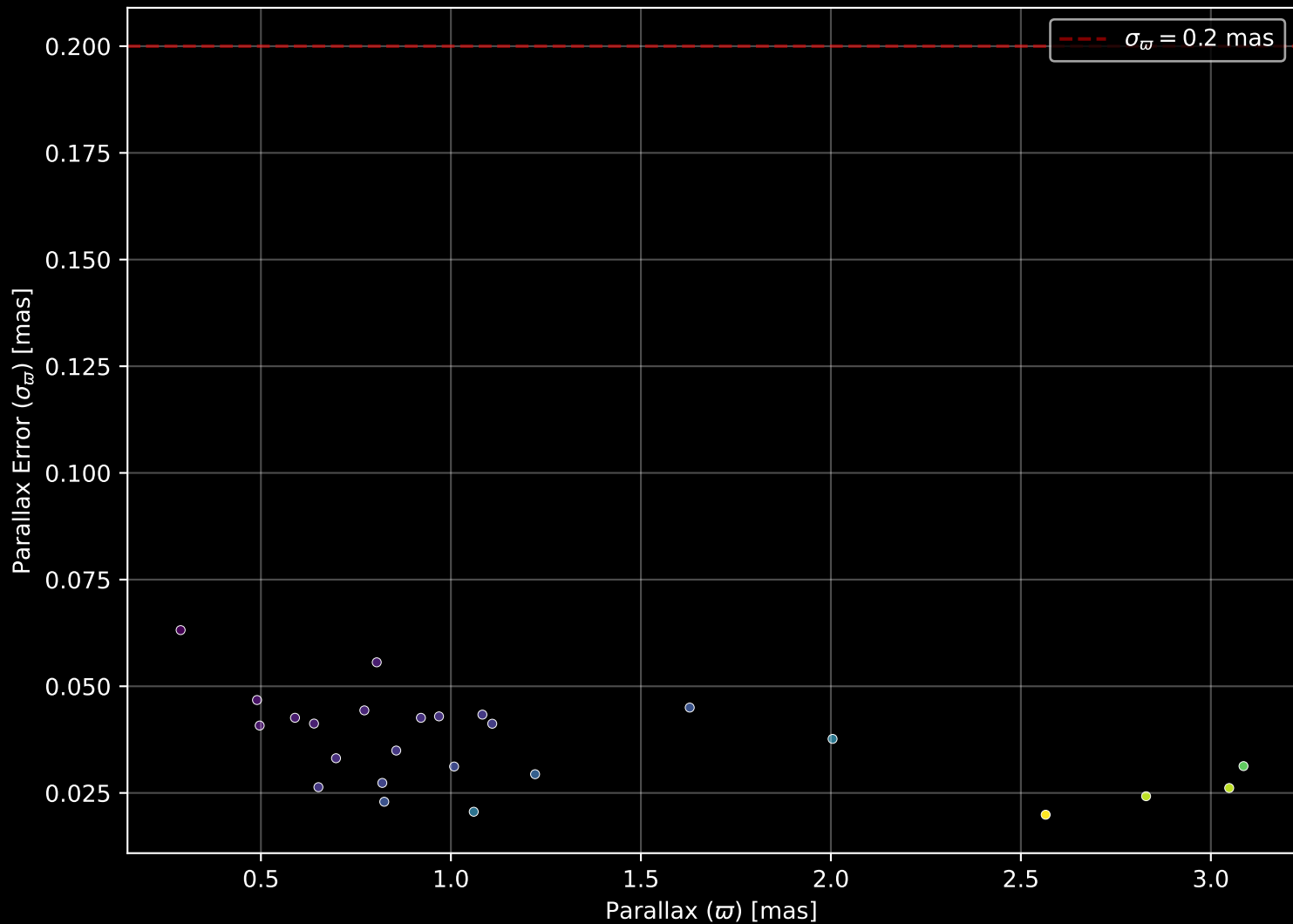
Cluster 50
[M/H] (mh_gspphot) (n=25)



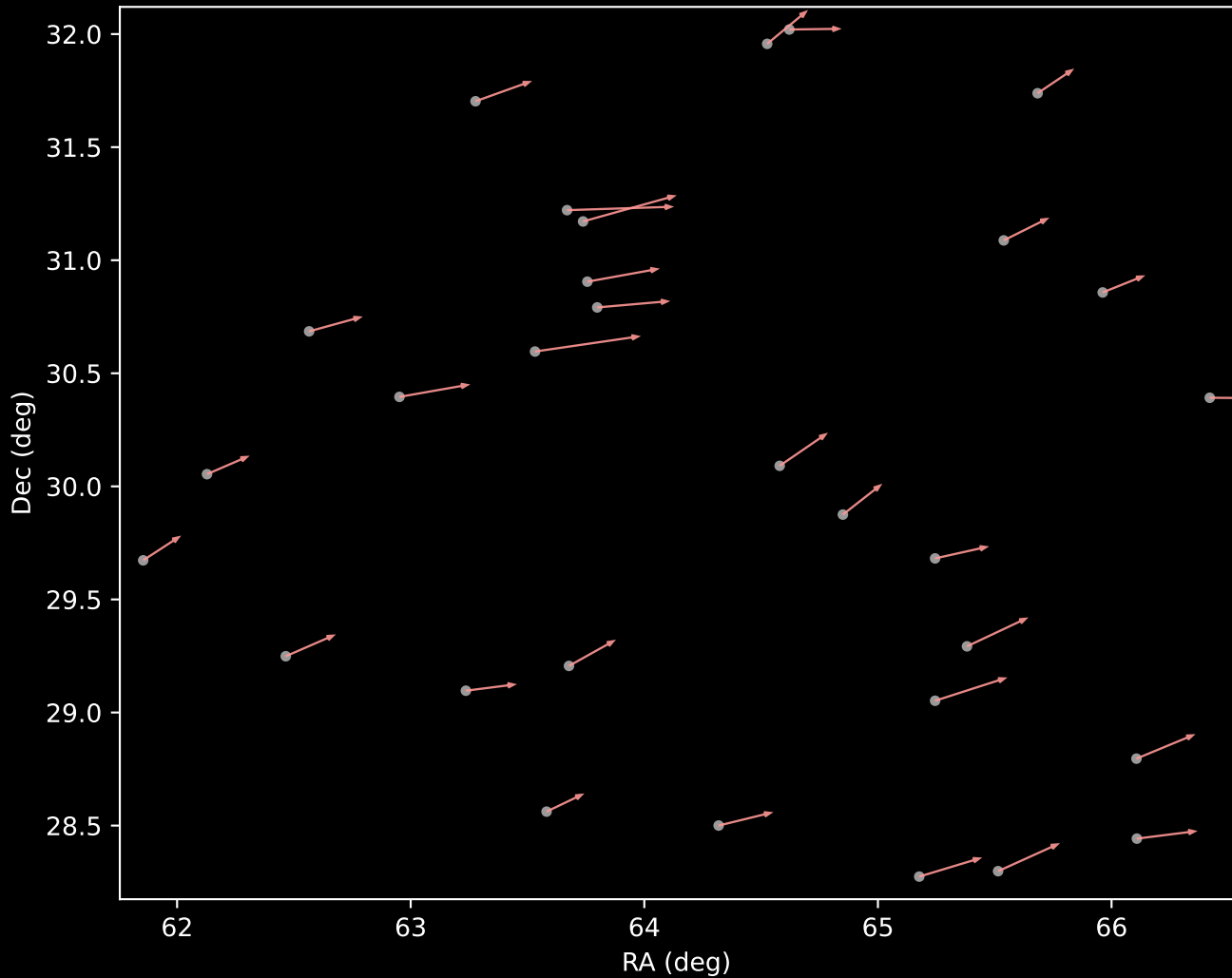
Cluster 50 Proper Motion Diagram (n=25)



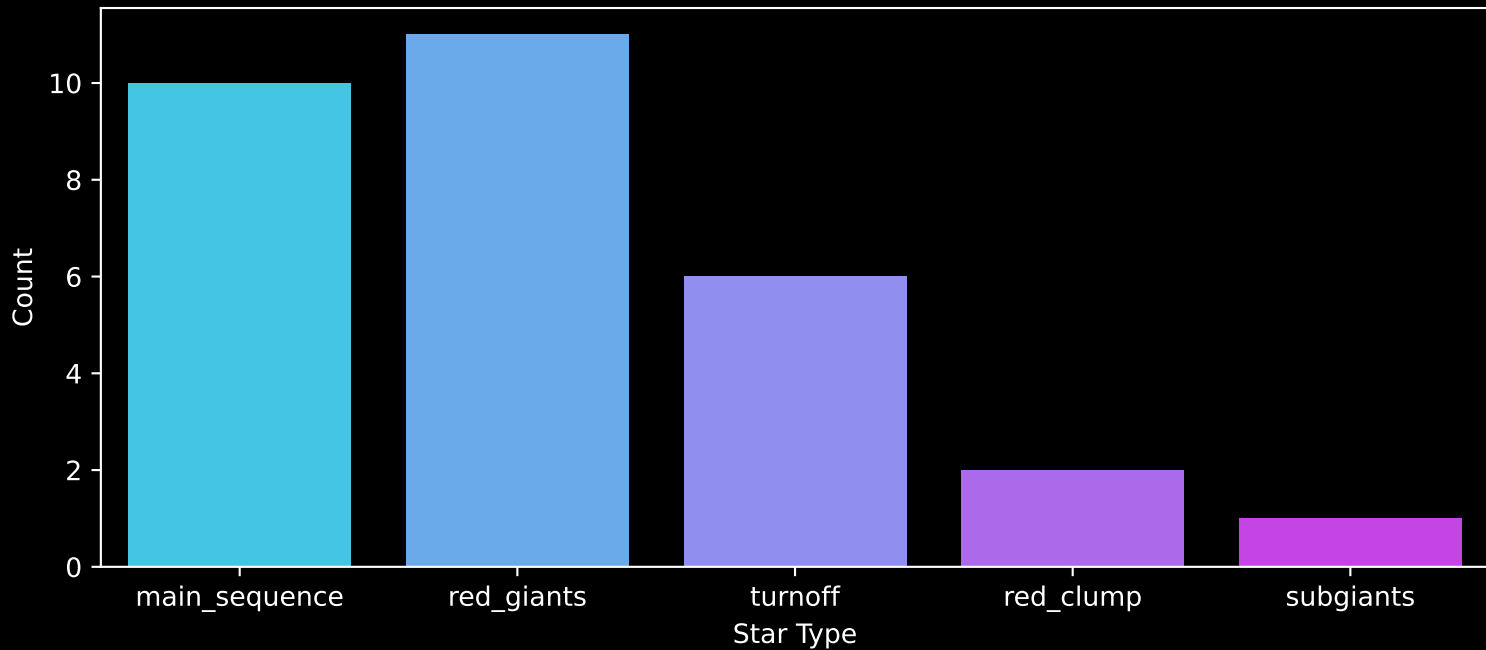
Cluster 50 Parallax Quality (n=25)



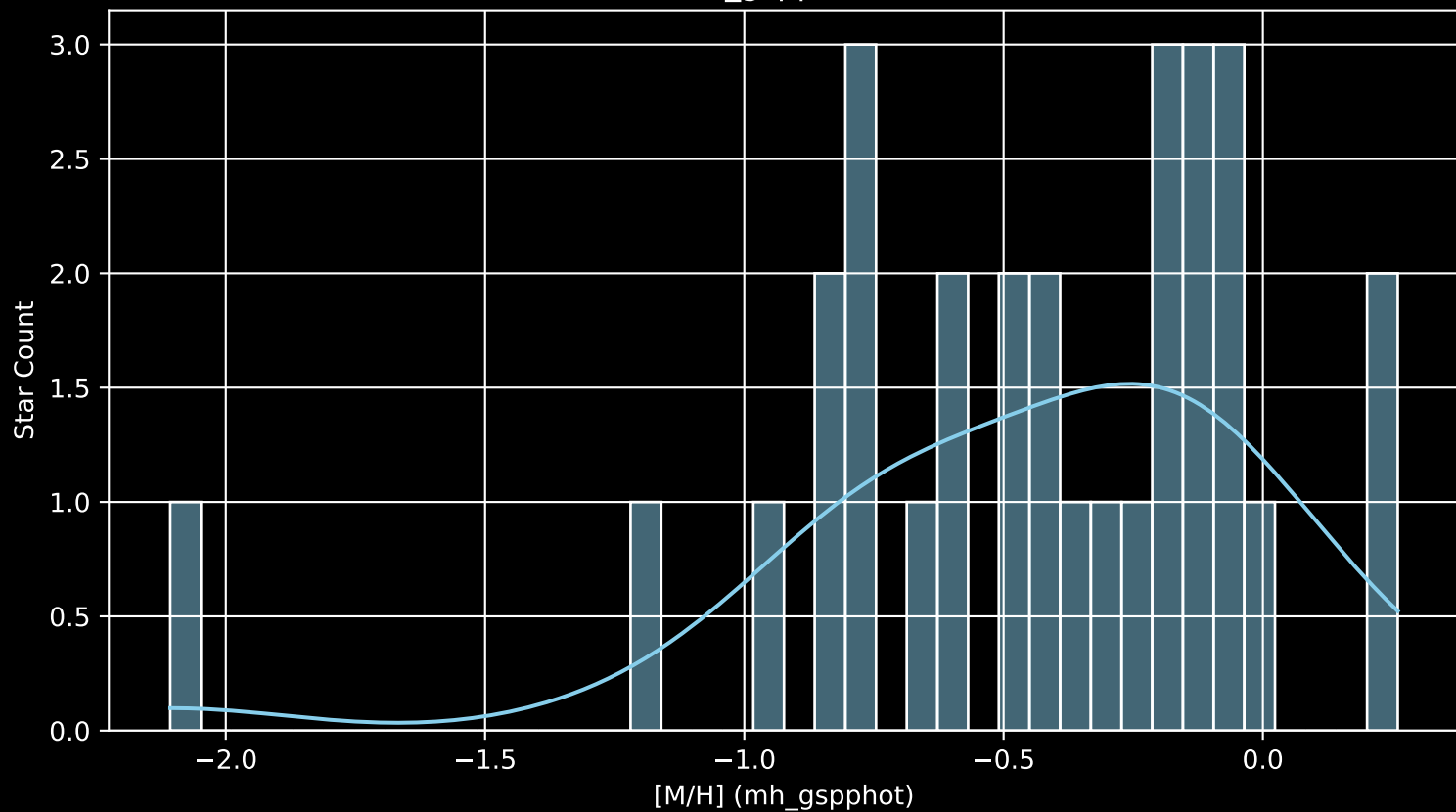
LAMOST DR11 Cluster 87 Proper Motion Vectors
(n=30)



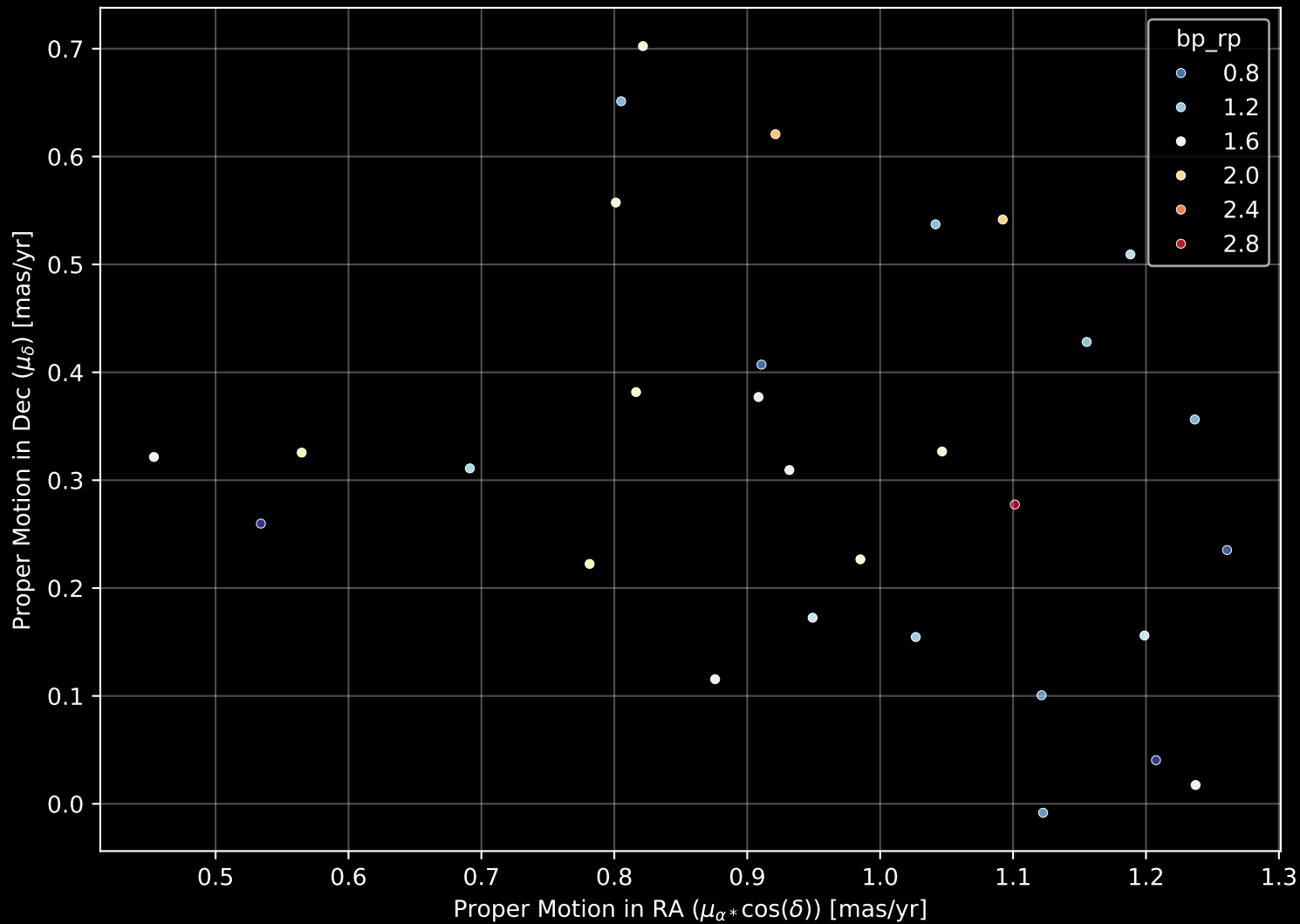
Cluster 87
Count plot of Star Type (n=30)



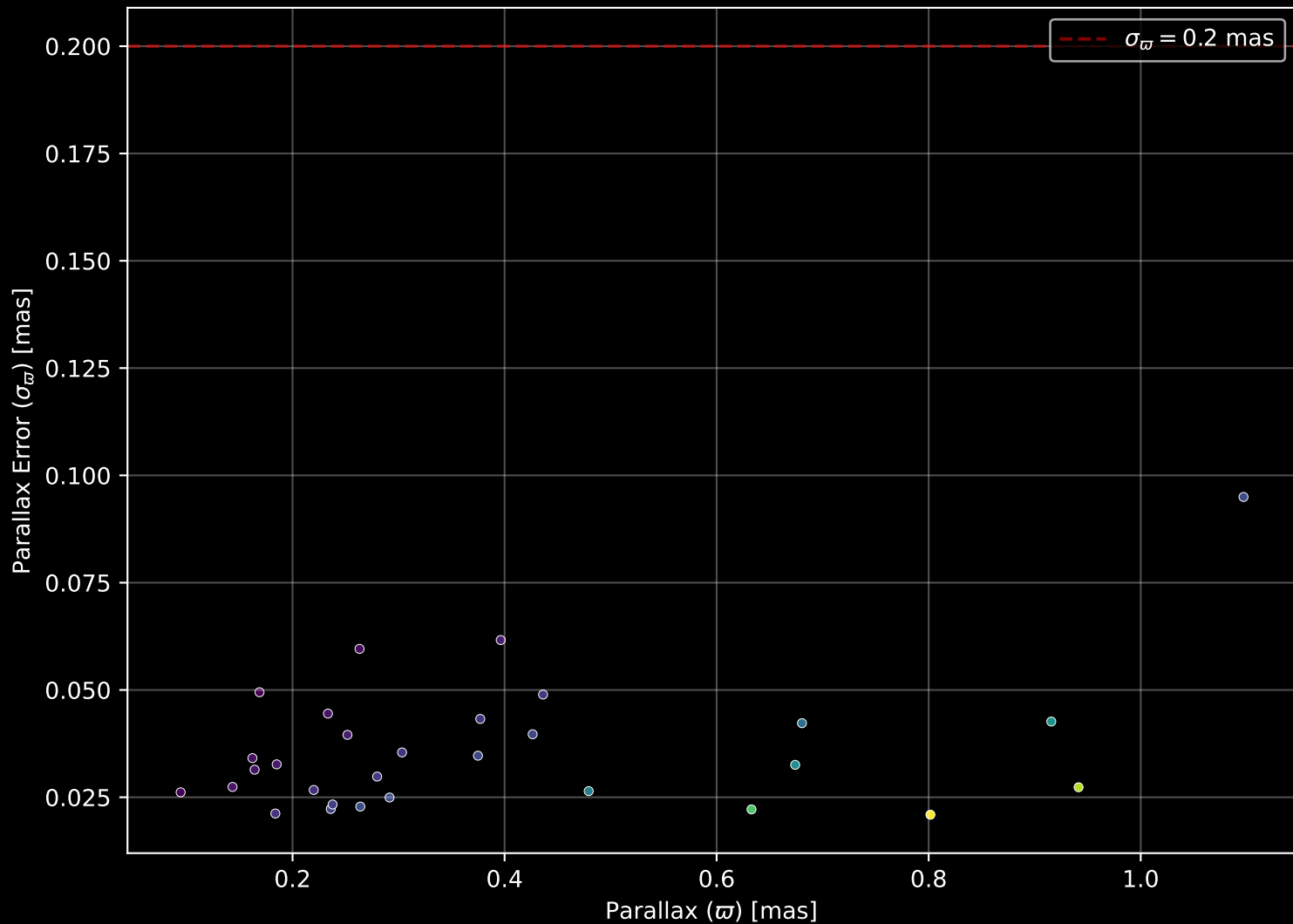
Cluster 87
[M/H] (mh_gspphot) (n=30)



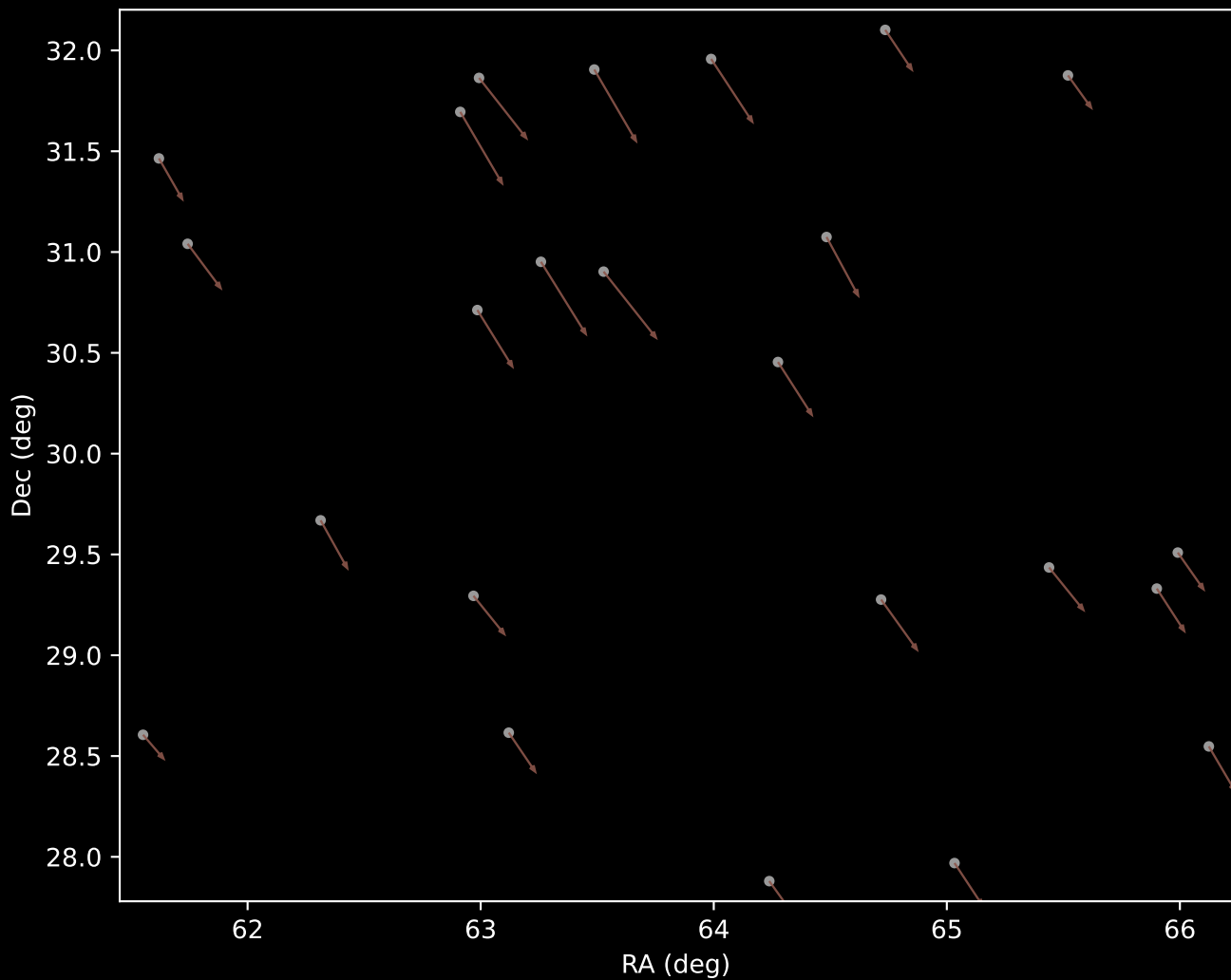
Cluster 87 Proper Motion Diagram (n=30)



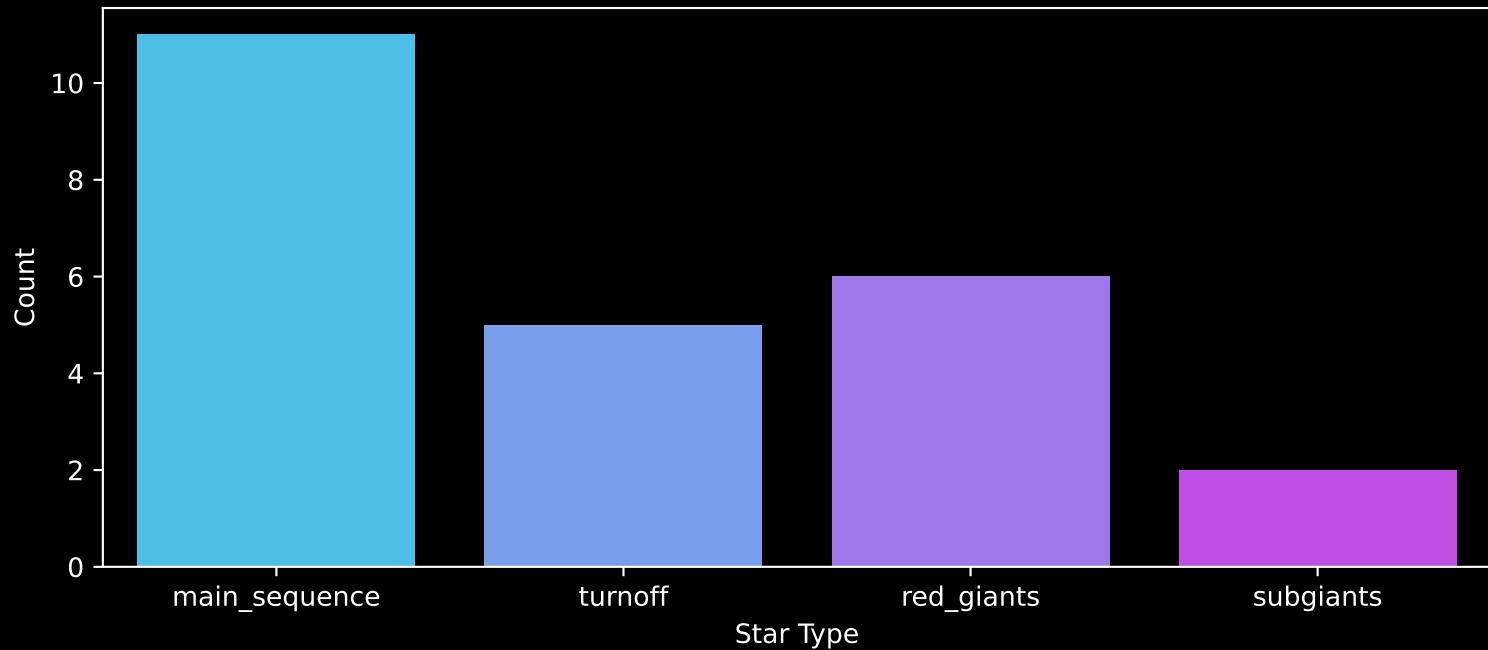
Cluster 87 Parallax Quality (n=30)



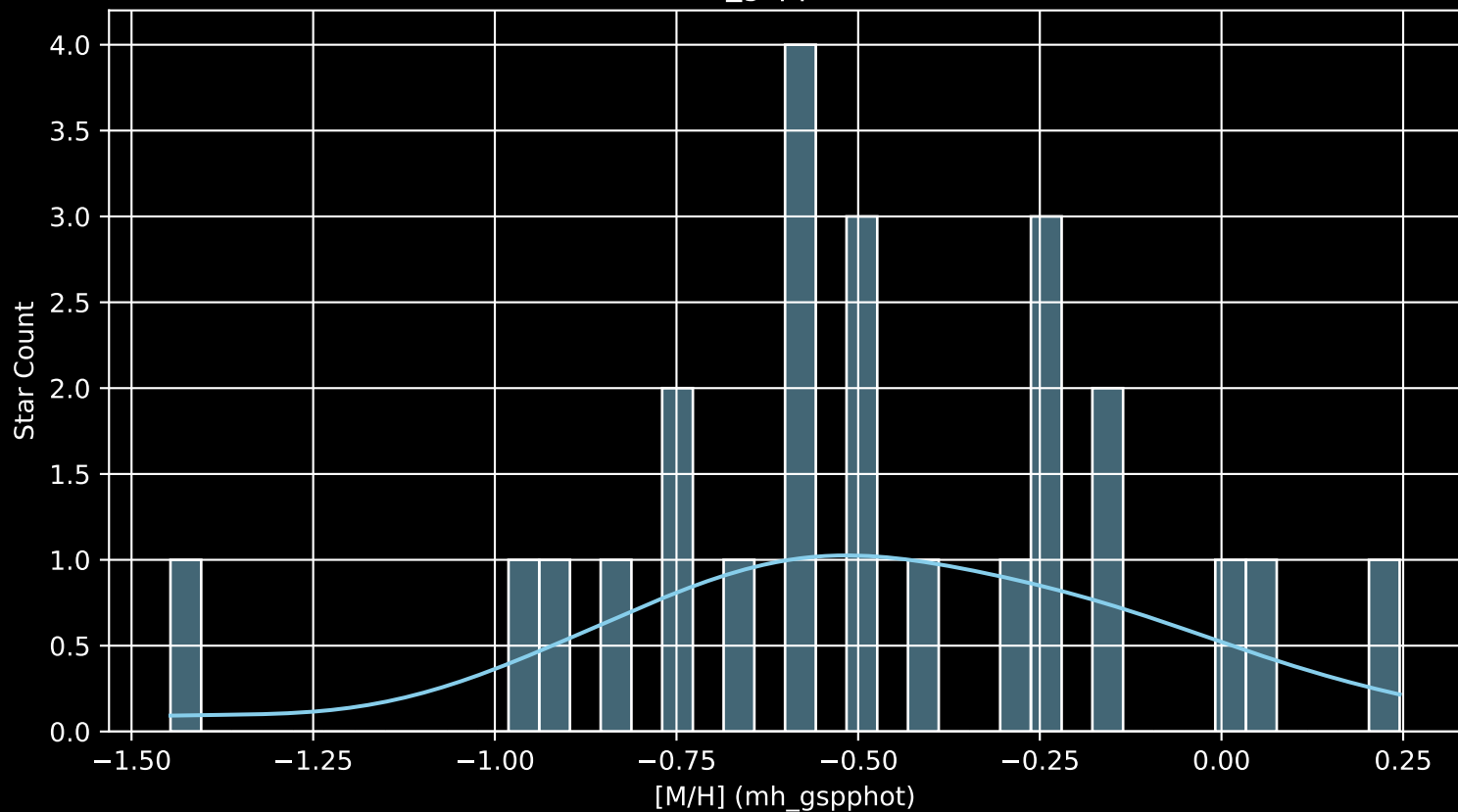
LAMOST DR11 Cluster 90 Proper Motion Vectors
(n=24)



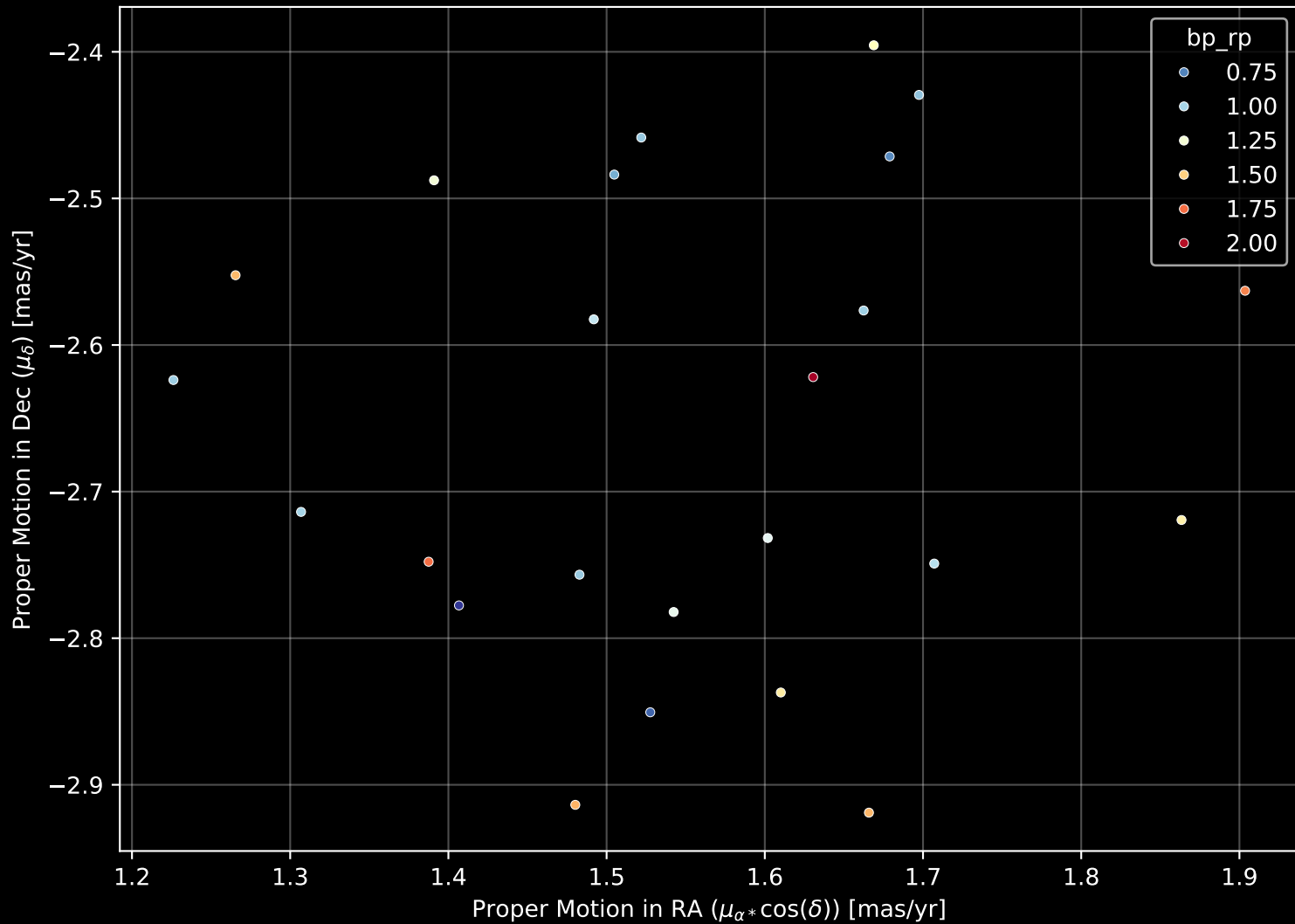
Cluster 90
Count plot of Star Type (n=24)



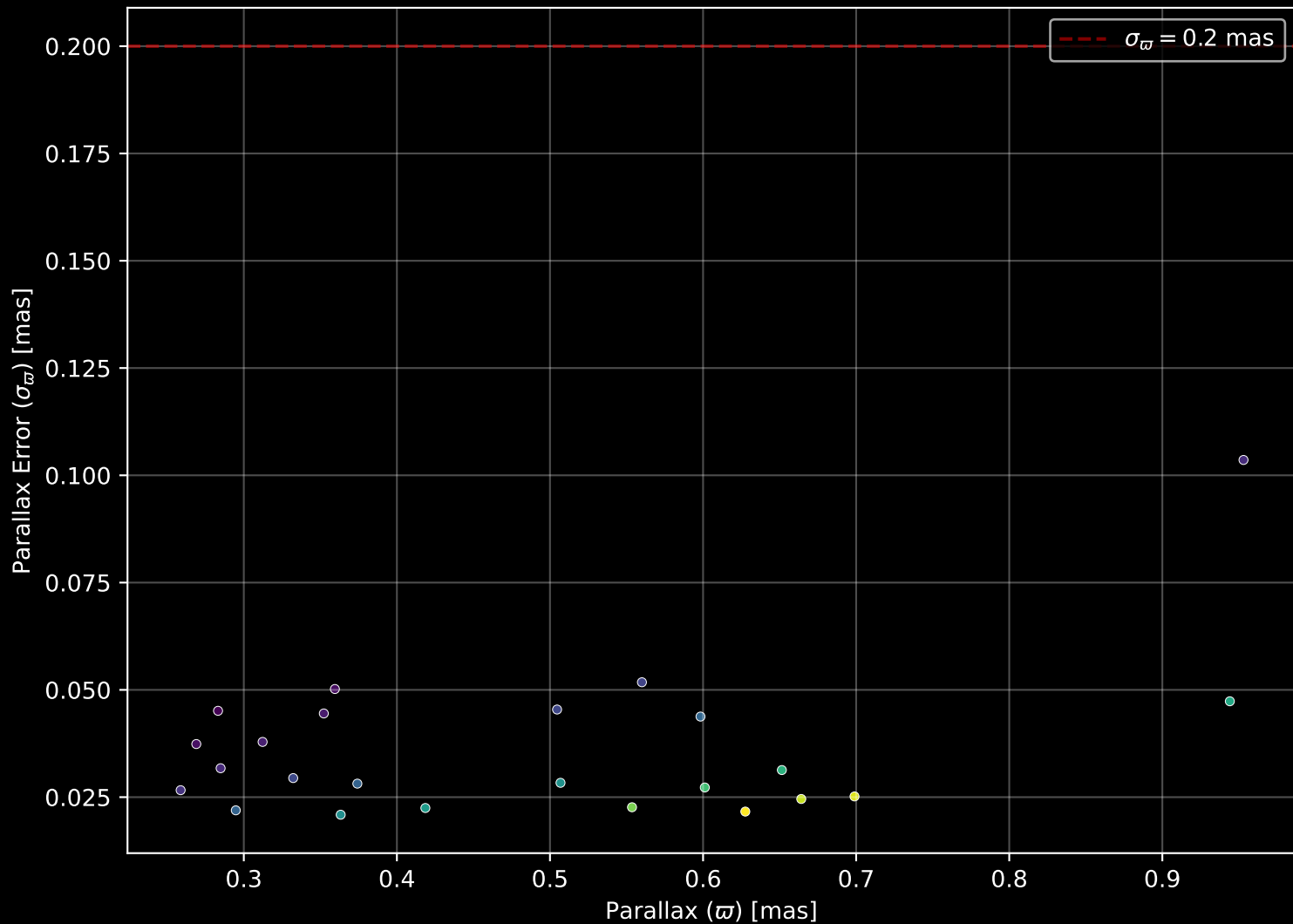
Cluster 90
[M/H] (mh_gspphot) (n=24)



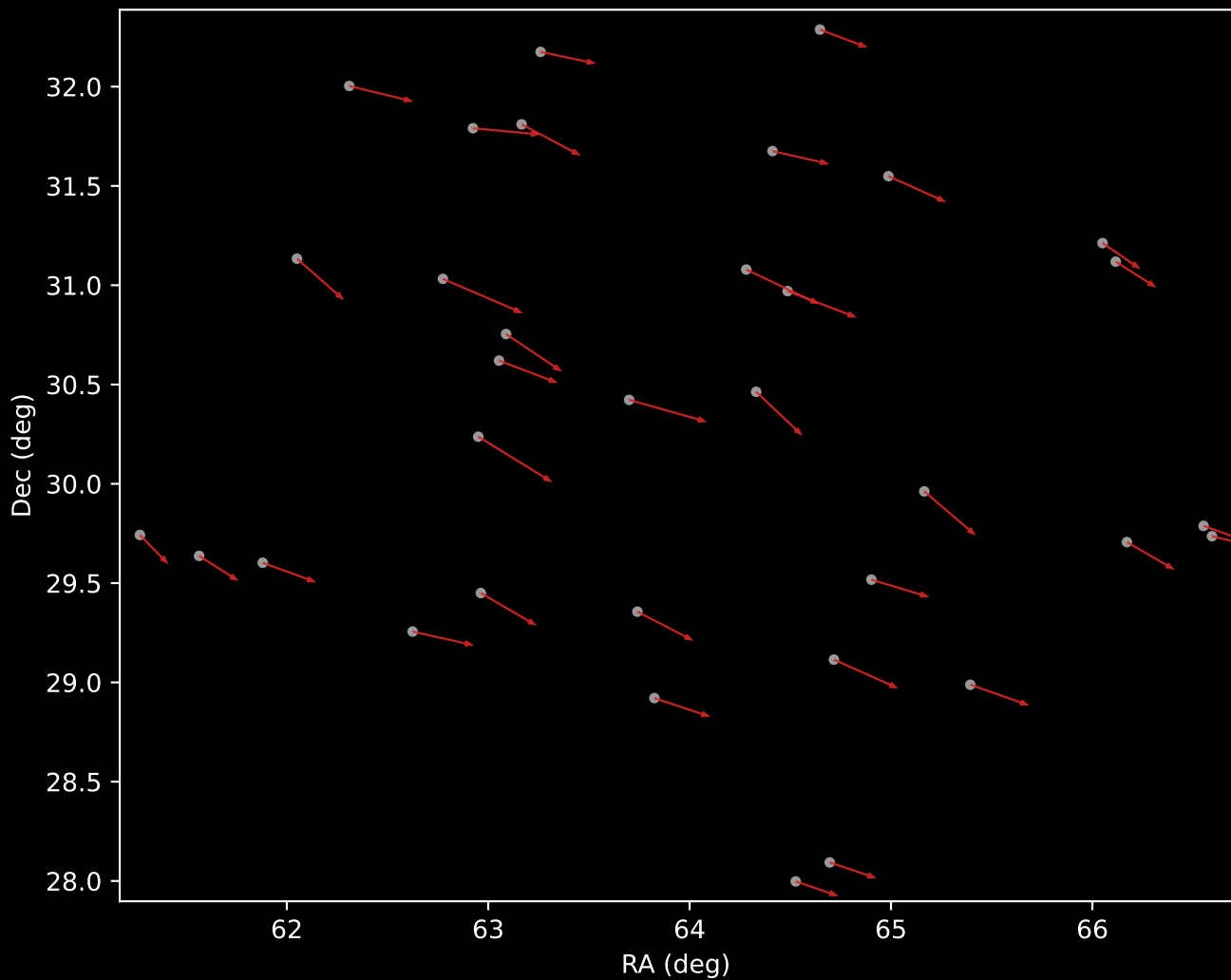
Cluster 90 Proper Motion Diagram (n=24)



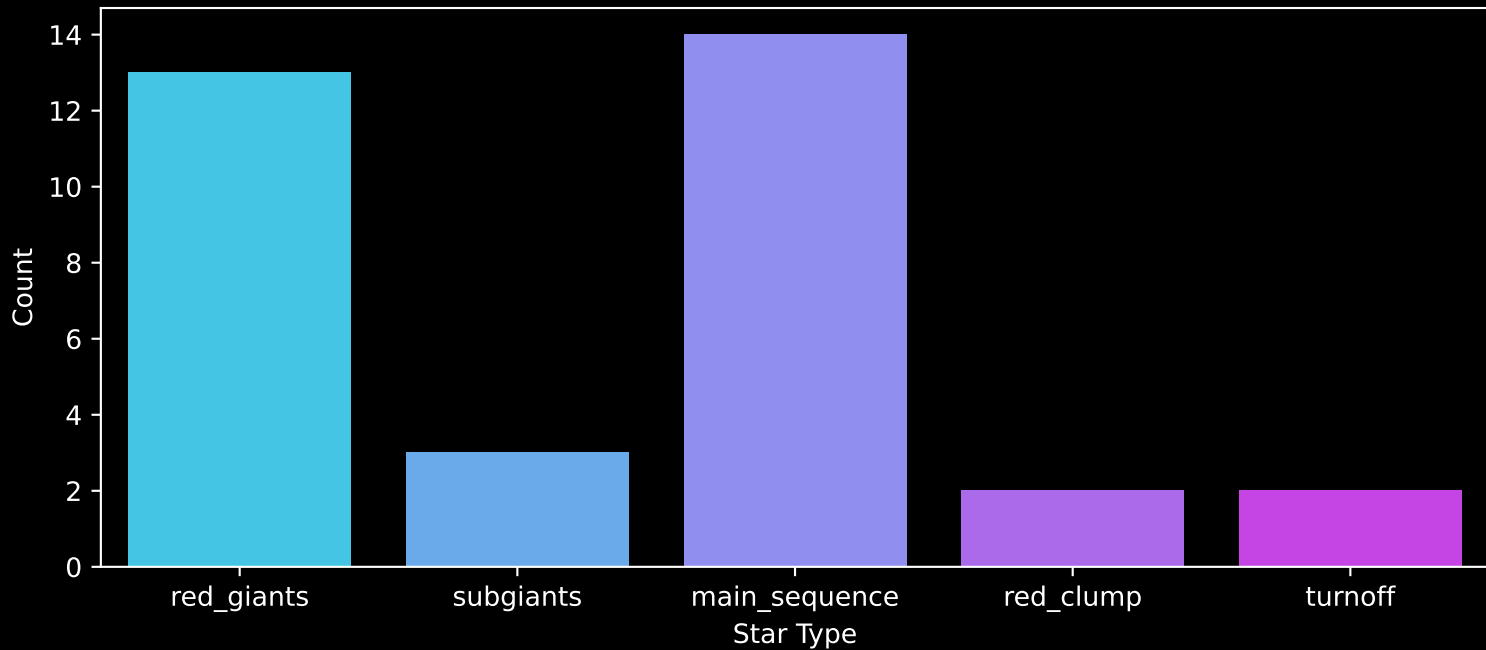
Cluster 90 Parallax Quality (n=24)



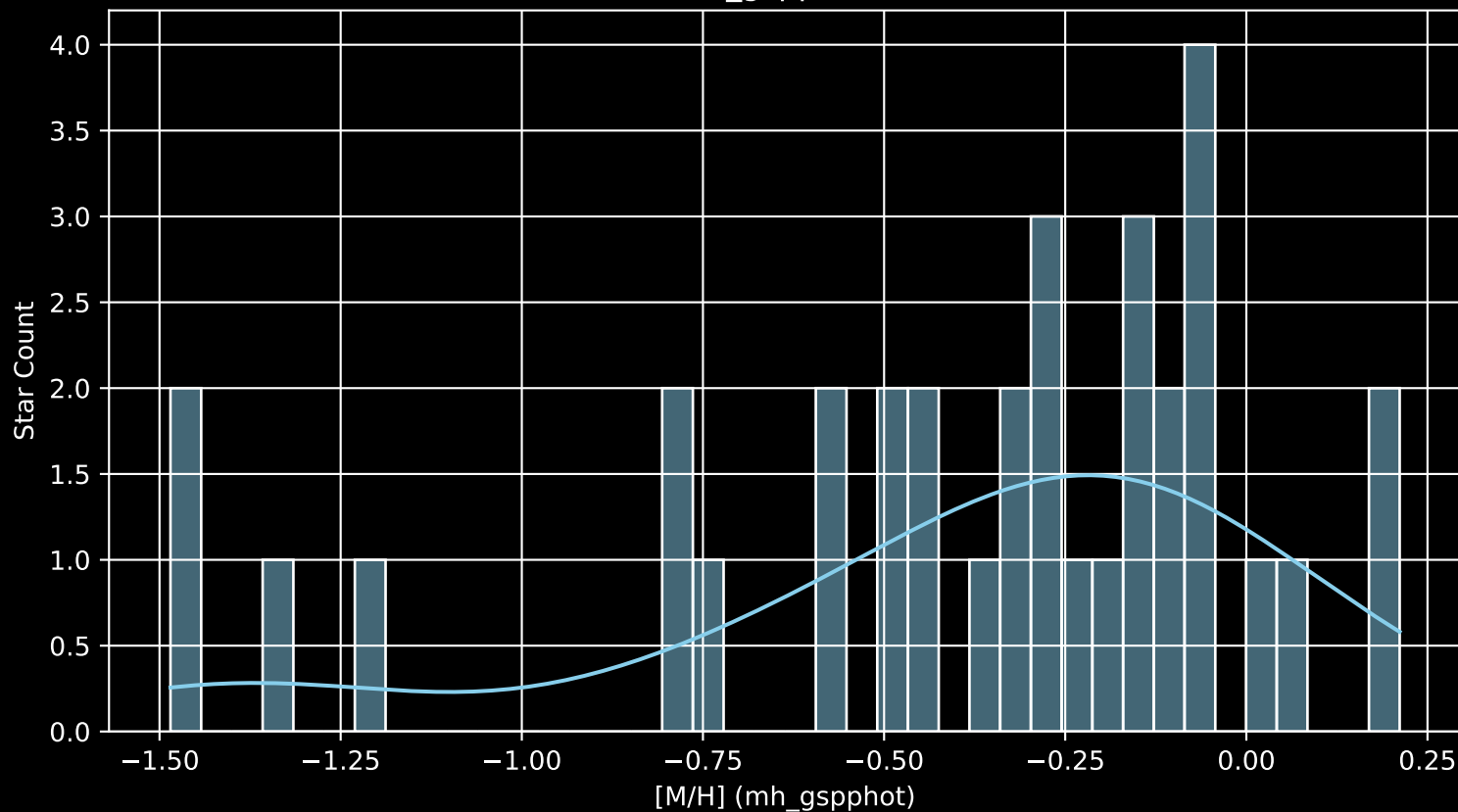
LAMOST DR11 Cluster 106 Proper Motion Vectors
(n=34)



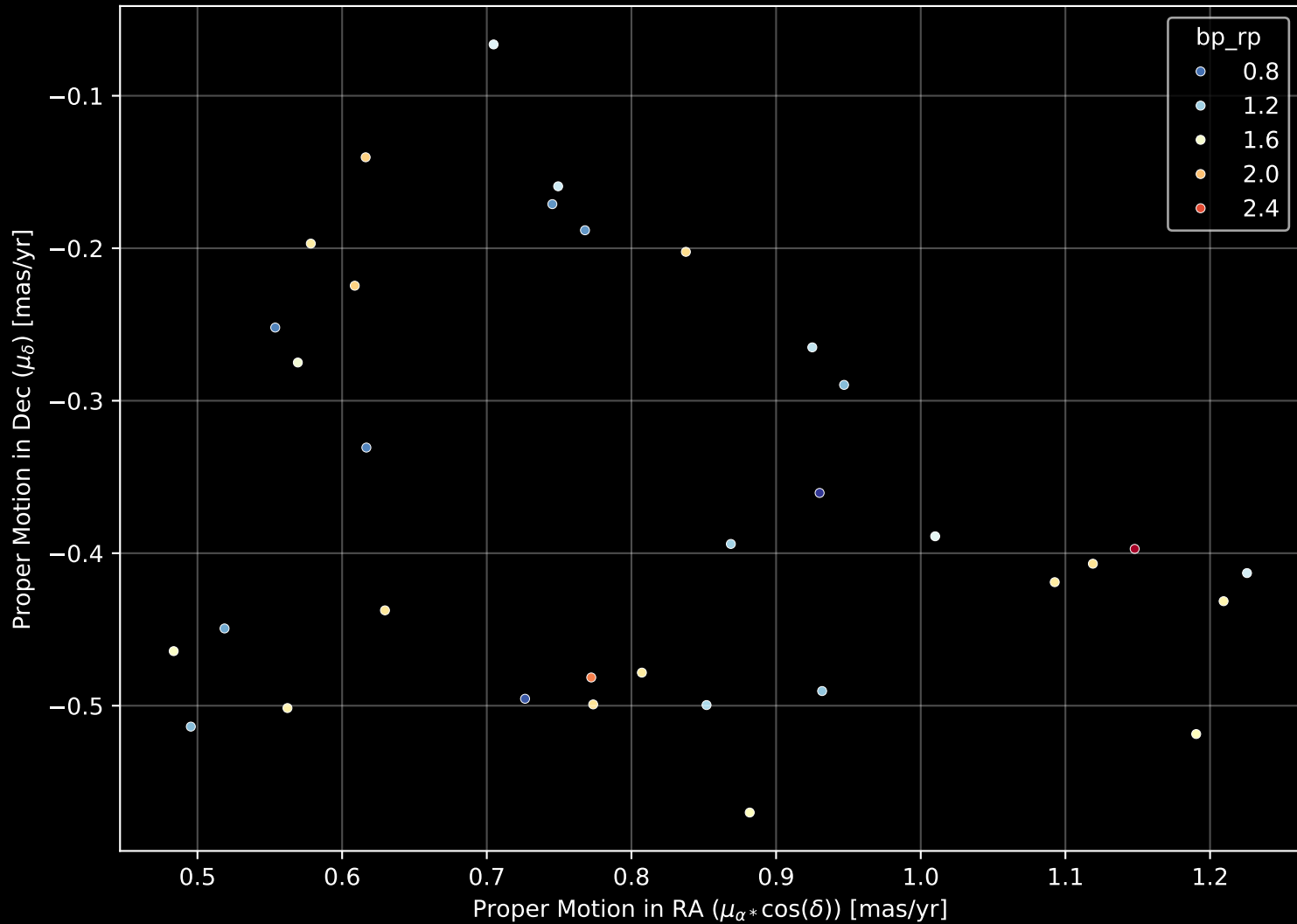
Cluster 106
Count plot of Star Type (n=34)



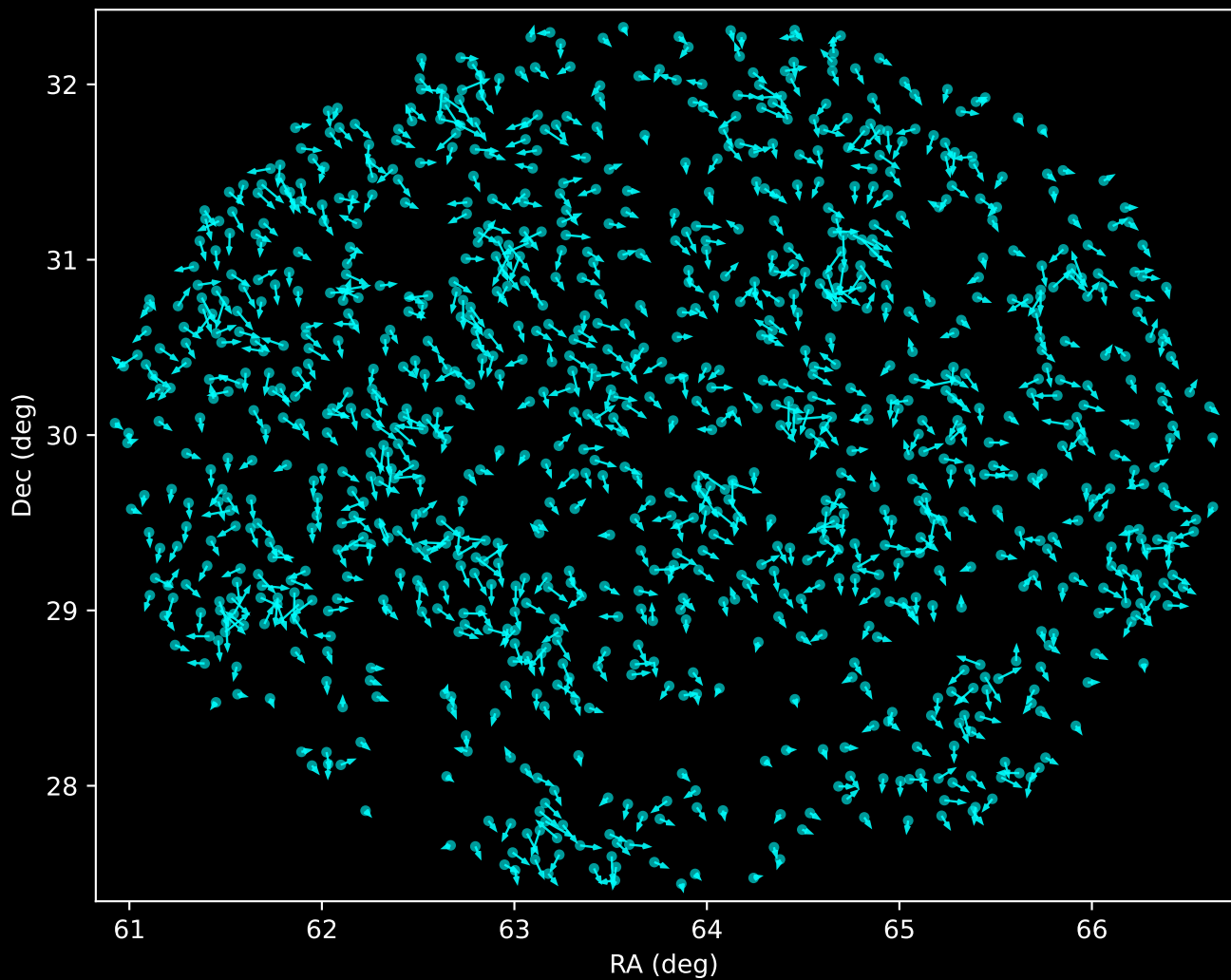
Cluster 106
[M/H] (mh_gspphot) (n=34)



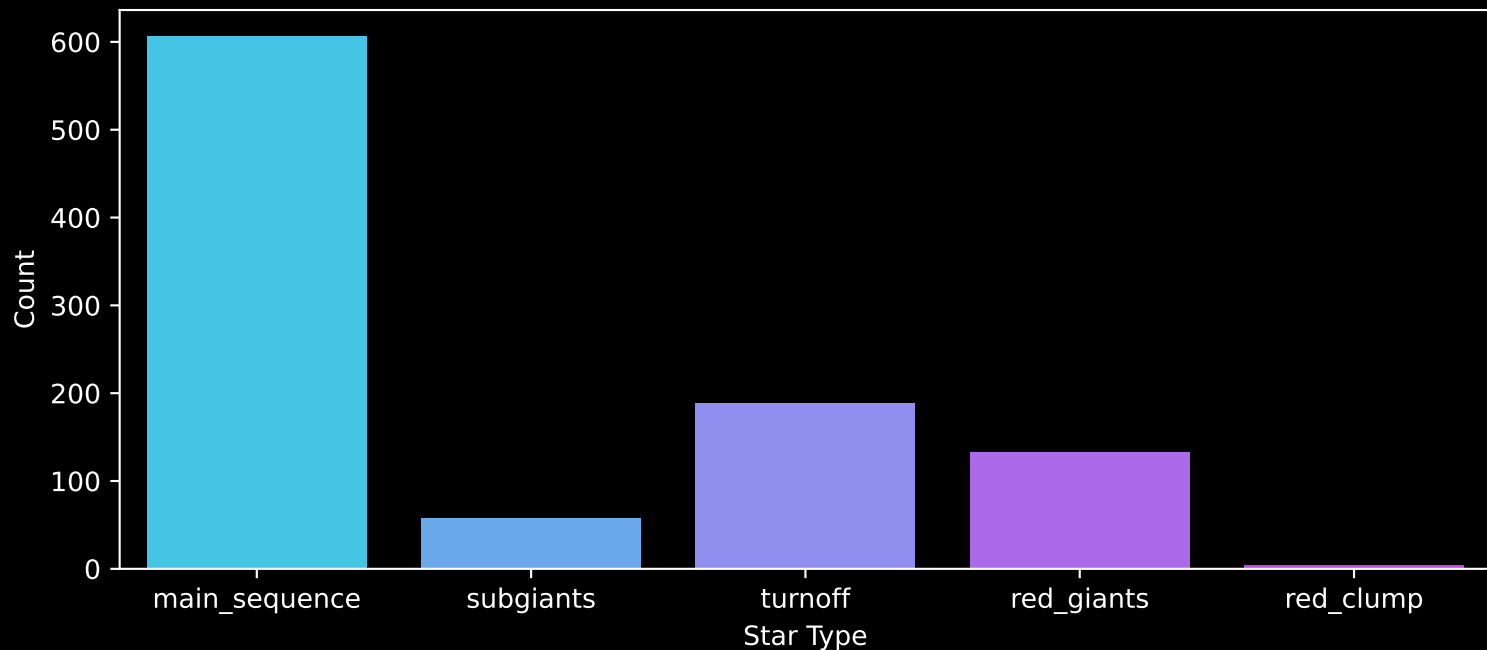
Cluster 106 Proper Motion Diagram (n=34)



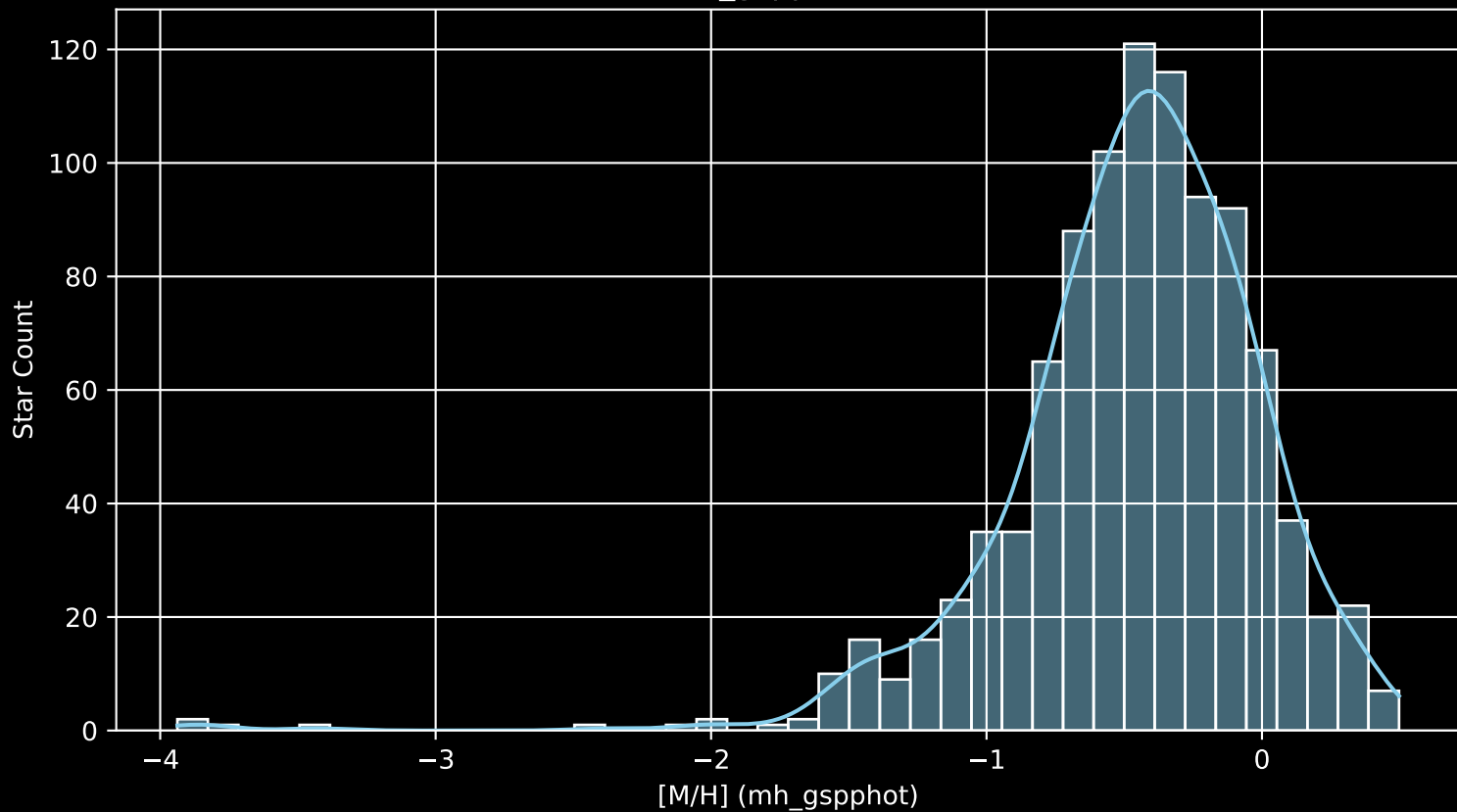
LAMOST DR11 Noise Stream Proper Motion Vectors
(n=989)



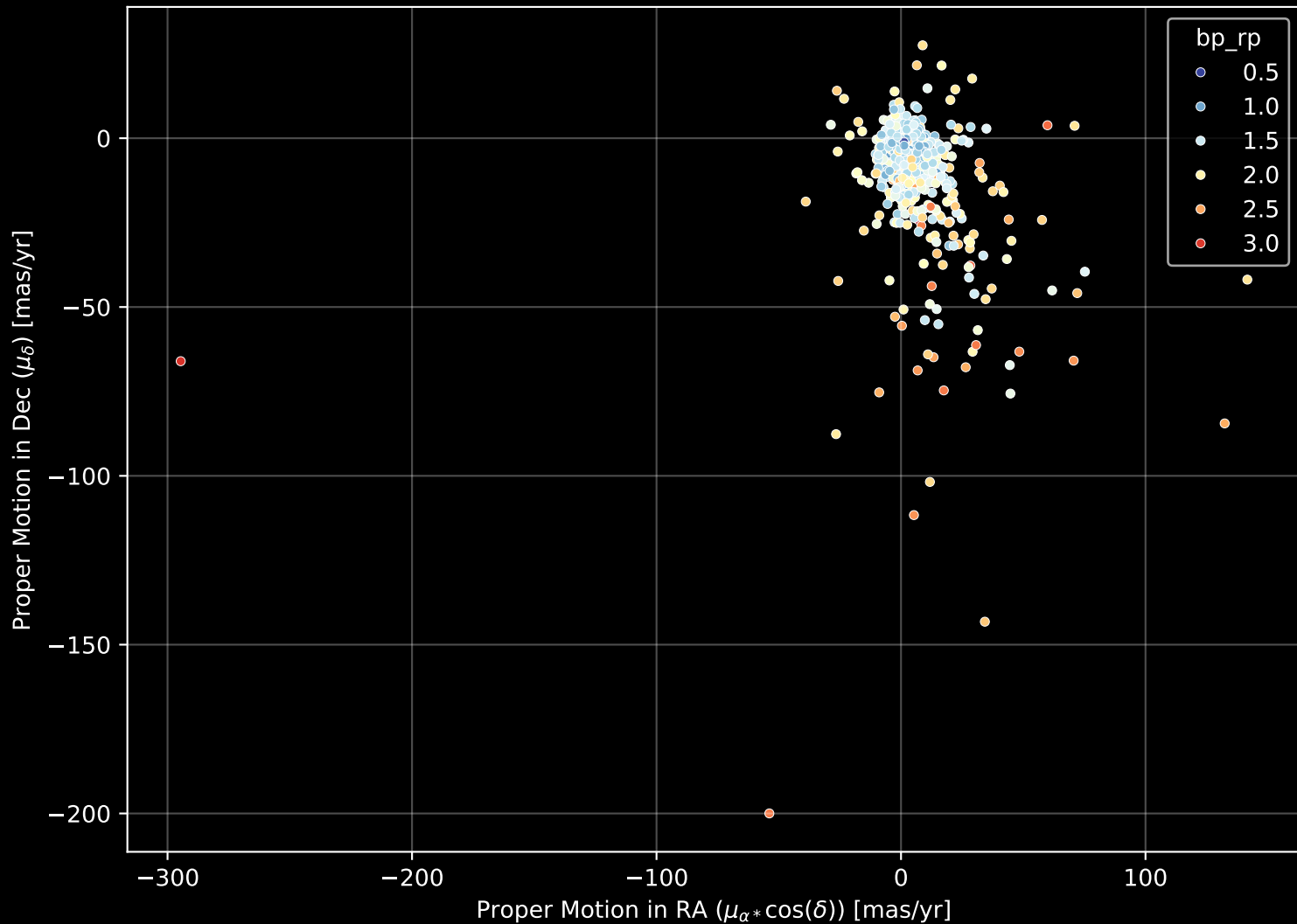
Cluster -1
Count plot of Star Type (n=989)



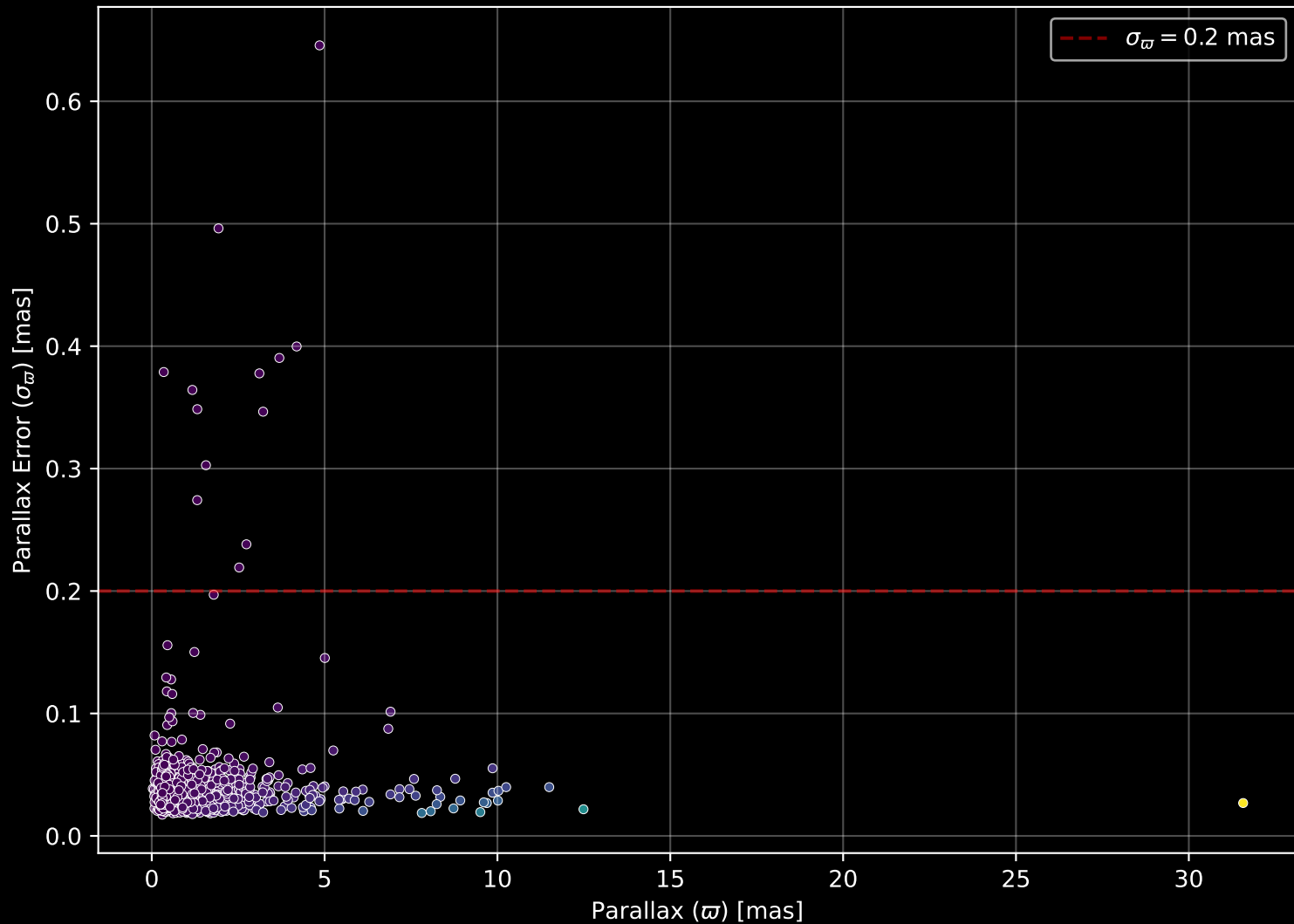
Cluster -1
[M/H] (mh_gspphot) (n=986)



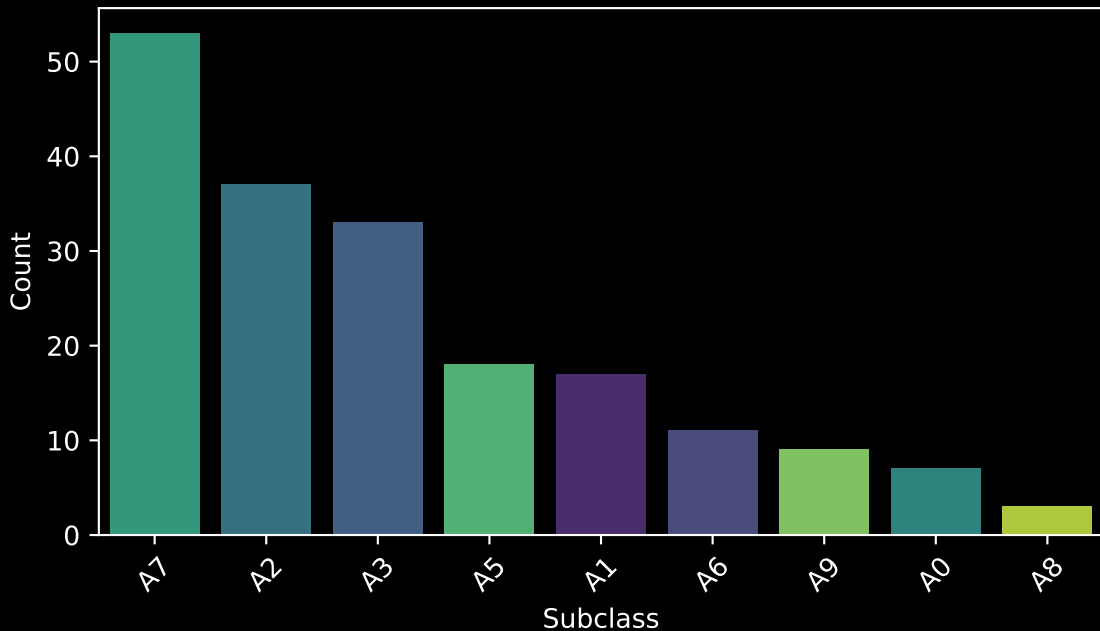
Cluster -1 Proper Motion Diagram (n=989)



Cluster -1 Parallax Quality (n=989)

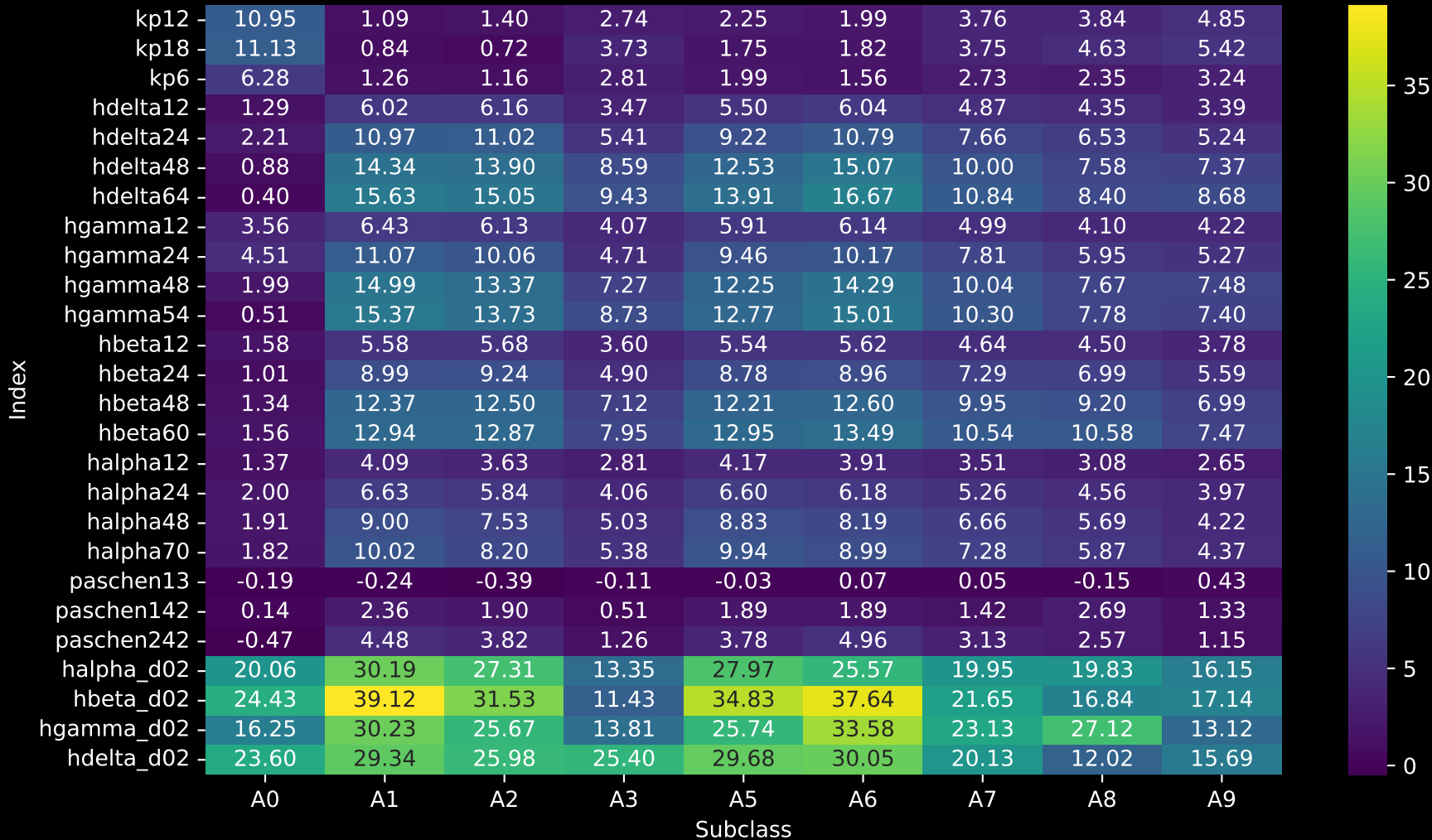


Subclass distribution for planid GAC_063N29_B1
(n= 188)

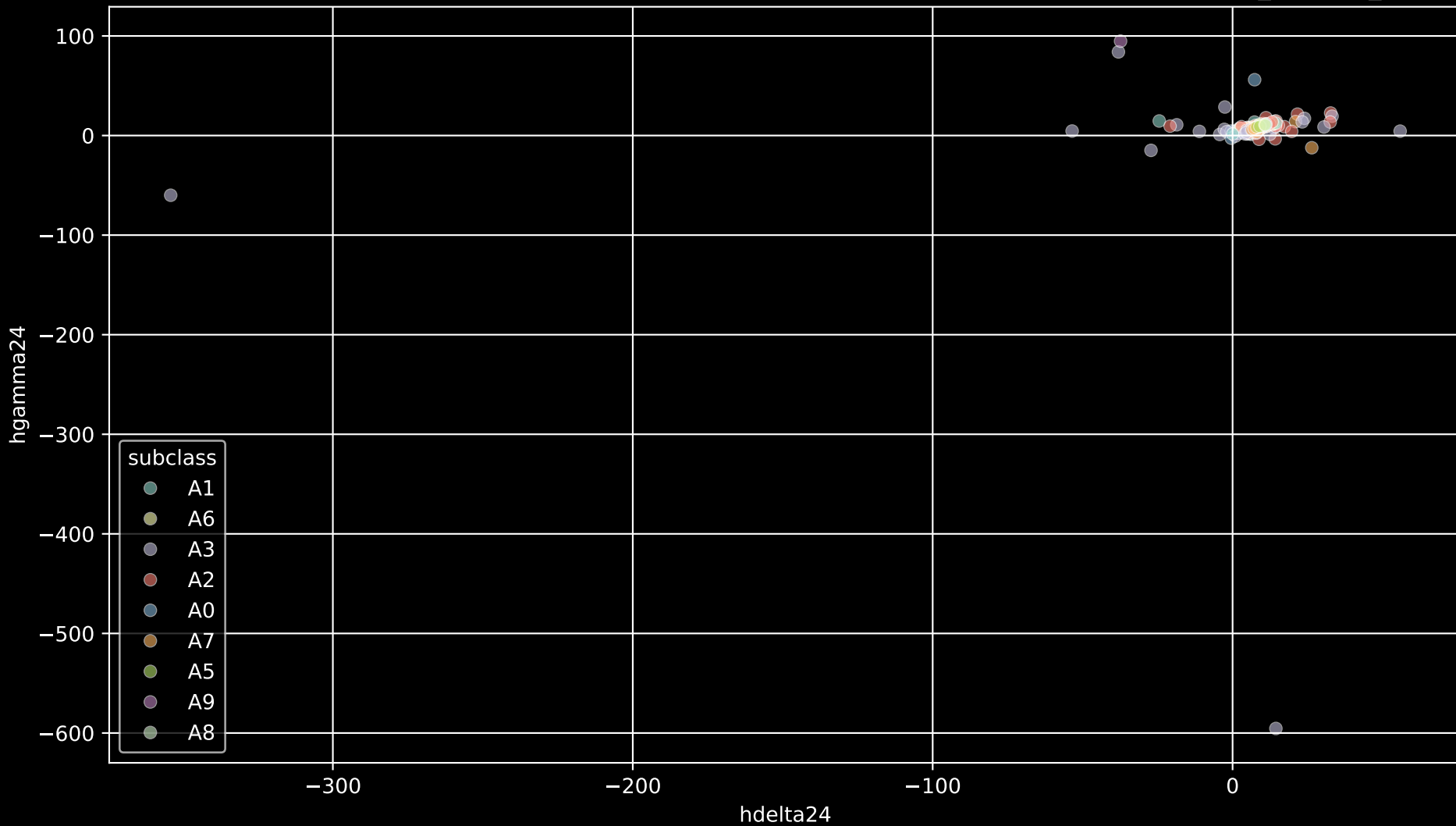


LAMOST DR11 Median Spectra Indices By Subclass

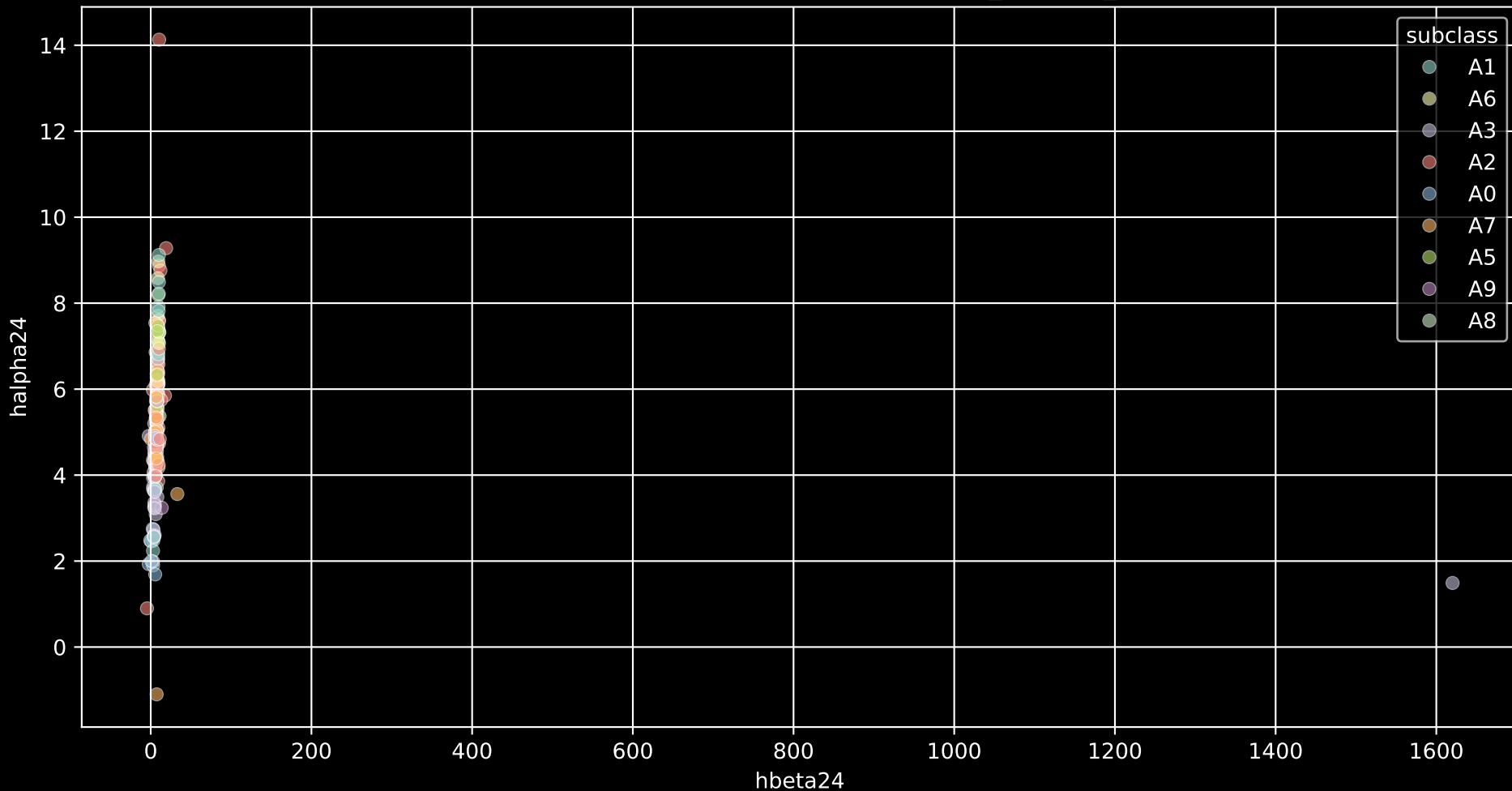
Star Field (planid) [GAC_063N29_B1]



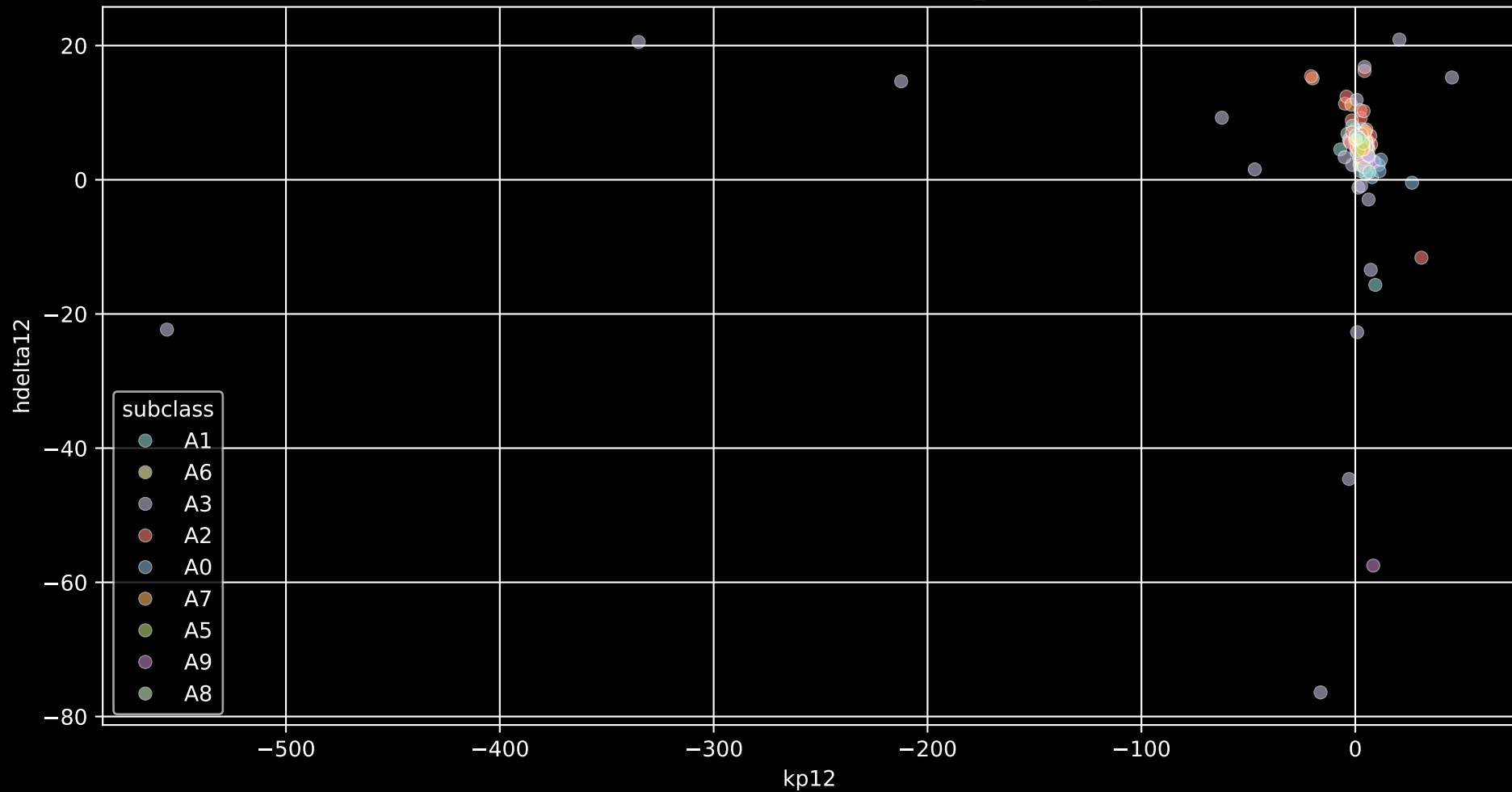
hdelta24 vs hgamma24 (H δ variant vs H γ variant; both Balmer diagnostics). (planid GAC_063N29_B1)



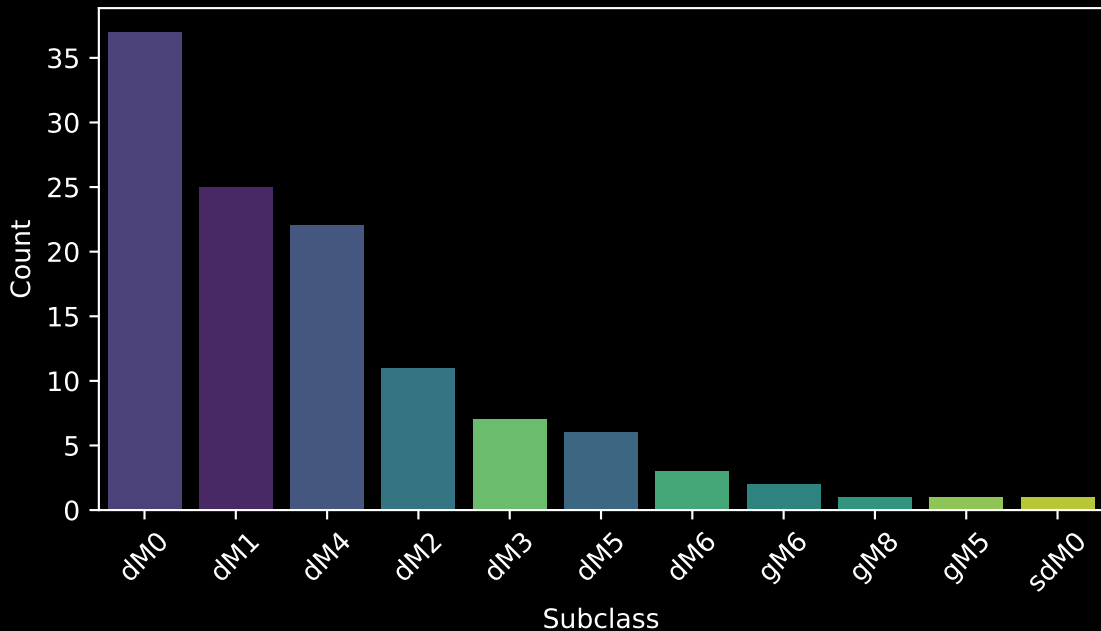
Expect a monotonic increase with cooling (A1 to A9),
but non-linear deviations highlight chromospheric activity or veiling (e.g., emission in H α for active stars)
A types: hbeta24 vs halpha24 (planid GAC_063N29_B1)



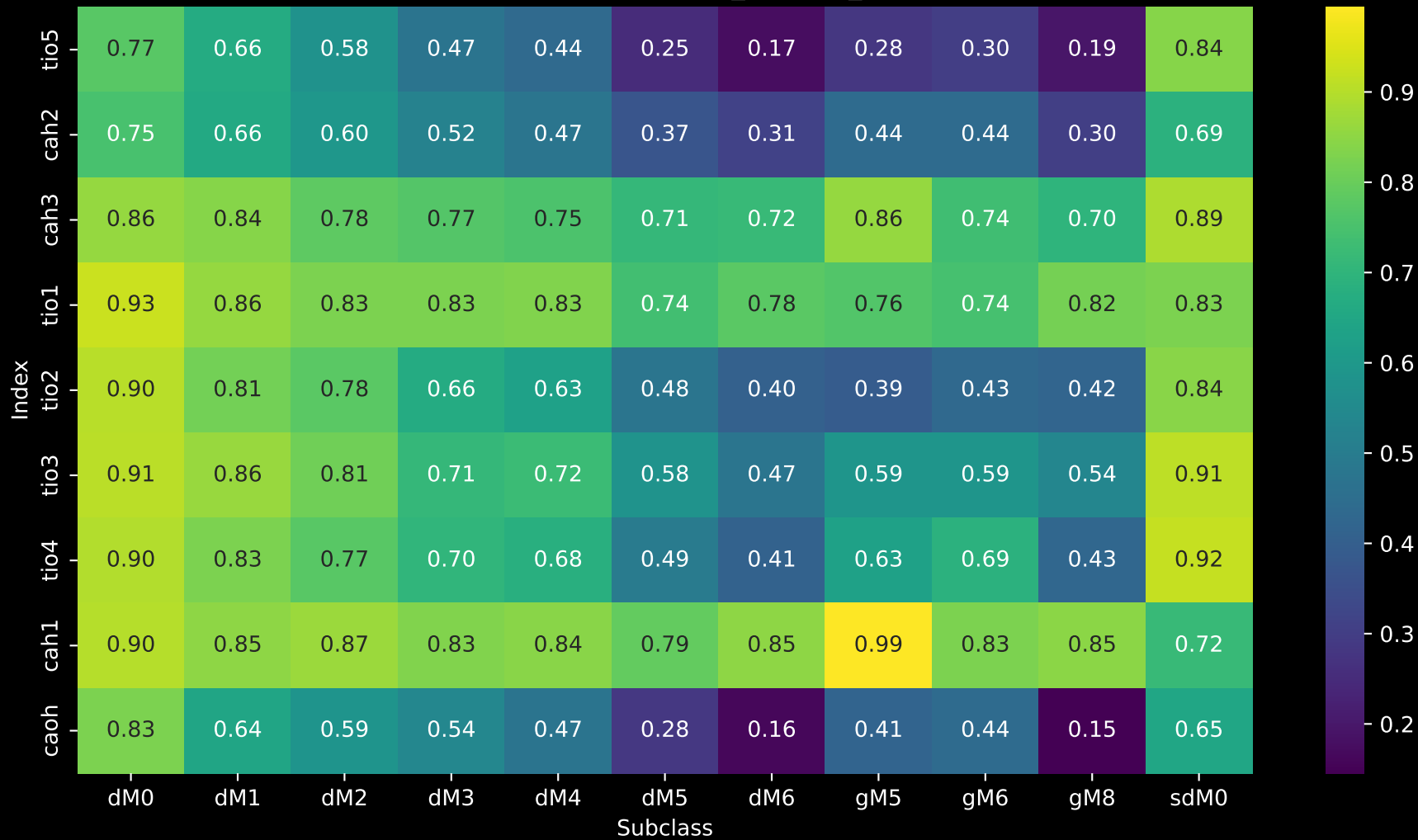
Metals vs Balmer would show anti-correlation in peculiar A stars (weaker Balmer in metal-rich).
For this field, it adds abundance insights, revealing if metal-poor halo interlopers (~ -3 [M/H] tail) affect A stars
A types: kp12 vs hdelta12 (planid GAC_063N29_B1)



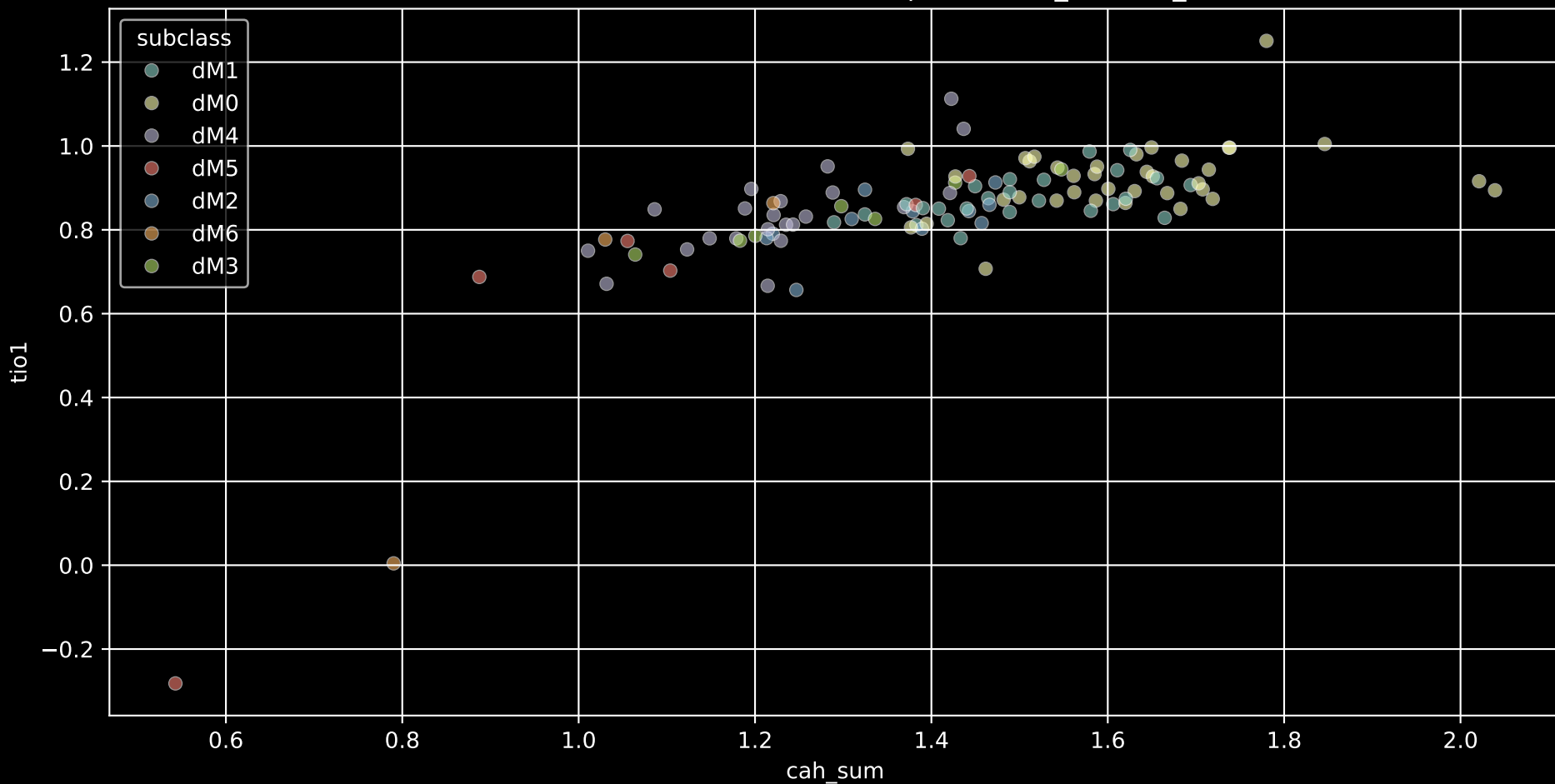
Subclass distribution for planid GAC_063N29_B1
(n= 116)



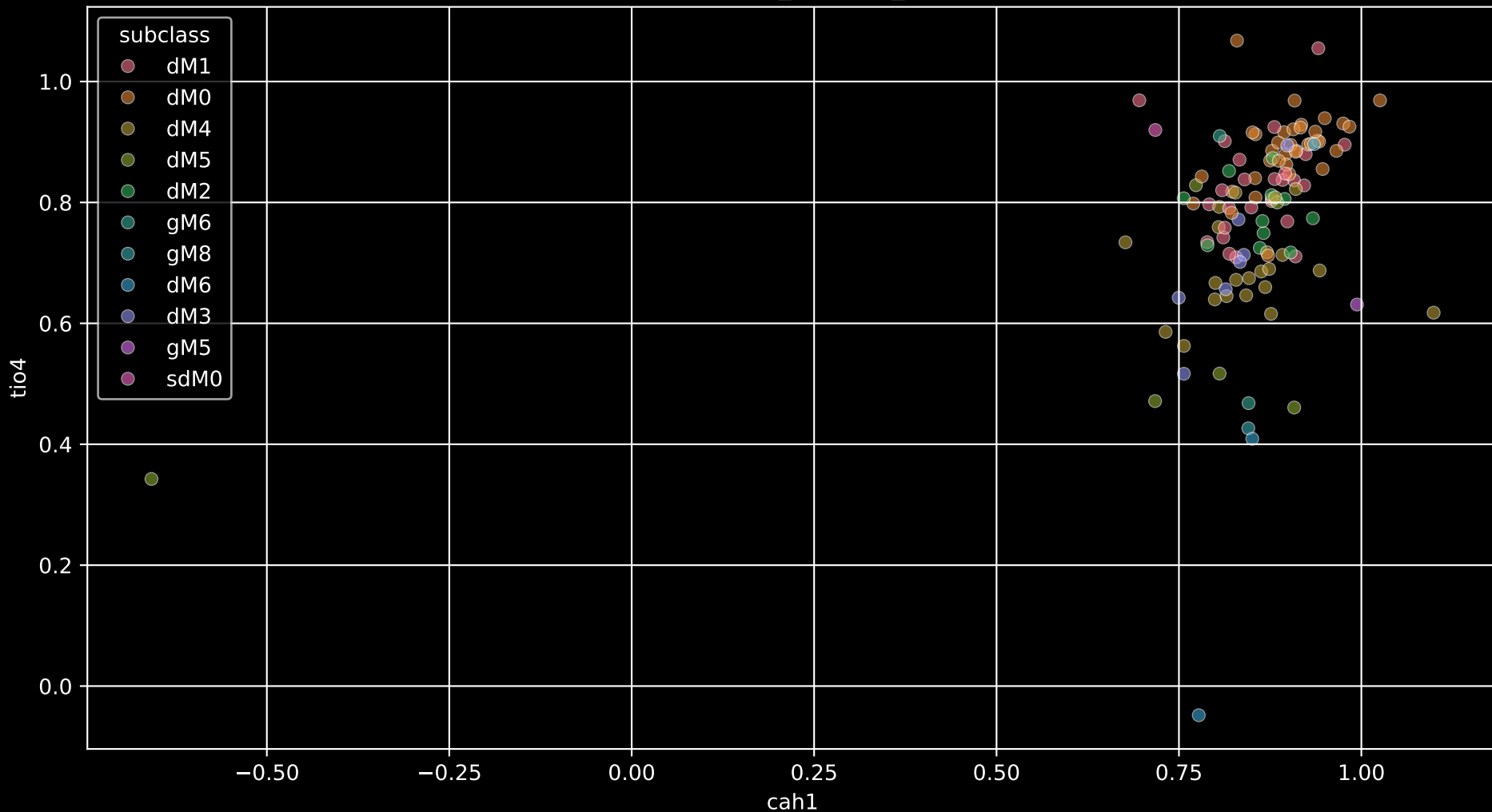
LAMOST DR11 Median Spectra Indices By Subclass
Star Field (planid) [GAC_063N29_B1]



This diagnostic diagram separates M dwarfs (strong CaH, moderate TiO) from giants (weaker CaH).
For this field, it would confirm all dM are dwarfs (no giant contamination in the ~ 7 Gyr median age population),
show anti-correlation with subtype (CaH weakens, TiO strengthens from dM0 to dM6)
(cah2 + cah3) vs tio1 (M dwarfs in planid GAC_063N29_B1)



cah1 vs tio4 (CaH variant vs another TiO; alternative for gravity/metallicity)
planid GAC_063N29_B1



cah1 vs tio2 (CaH variant vs TiO; for abundance vs temperature)
planid GAC_063N29_B1

